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METHODS, SYSTEMS, AND APPARATUSES FOR OPERATING A CONVEYOR

Abstract

Various embodiments described herein relate to management of a conveyor system. In this regard, a conveyor system includes a first set of slave rollers defining a first conveying section and a second set of slave rollers defining a second conveying section separate from the first conveying section. The conveyor system further includes a motor driven roller including at least a fixed pulley, a first independent pulley, and a second independent pulley. In addition, the first independent pulley is coupled with the first set of slave rollers and the second independent pulley is coupled with the second set of slave rollers. Moreover, the motor driven roller is configured to drive at least one of the first set of slave rollers or the second set of slave rollers based on selective engagement of at least one of the first independent pulley or the second independent pulley.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority pursuant to 35 U.S.C. 119 (a) to Indian Application No. 202411015967, filed Mar. 6, 2024, which application is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to a conveyor, and more particularly to methods, systems, and apparatuses for operating a conveyor.

BACKGROUND

[0003] Generally, a conveyer may include a series of rollers that may rotate to traverse a load, for example boxes, supplies, materials, other kind of objects from one location to another location. Usually, conveyor includes multiple zones that may be controlled independently to traverse the load from one location to another. Typically, independently controlling the various zones of the conveyor may require independent electrical drive systems and control systems that may add to the cost of the overall conveyor.

[0004] However, the applicant has identified many technical challenges and difficulties associated with conventional conveyor. Through applied effort, ingenuity, and innovation, the Applicant has solved problems relating to conveyer by developing solutions embodied in the present disclosure, which are described in detail below.

SUMMARY

[0005] The present disclosure provides a conveyor system comprising a first set of slave rollers defining a first conveying section; a second set of slave rollers defining a second conveying section separate from the first conveying section; and a motor driver roller (MDR) comprising at least a fixed pulley, a first independent pulley, and a second independent pulley, wherein the first independent pulley is coupled with the first set of slave rollers and the second independent pulley is coupled with the second set of slave rollers, and wherein the MDR is configured to drive at least one of the first set of slave rollers or the second set of slave rollers based on selective engagement of at least one of the first independent pulley or the second independent pulley.

[0006] The present disclosure further provides a method of controlling a conveyor system, comprising: defining a first conveying section by a first set of slave rollers; defining a second conveying section by a second set of slave rollers, wherein the second conveying section is defined separately from the first conveying section; arranging a motor driven roller (MDR) between the first conveying section and the second conveying section, wherein the MDR comprising at least a fixed pulley, a first independent pulley, and a second independent pulley; coupling the first independent pulley of the MDR with the first set of slave rollers; coupling the second independent pulley of the MDR with the second set of slave rollers; selectively engaging at least one of the first independent pulley or the second independent pulley with the fixed pulley of the MDR; and driving, through the MDR, at least one of the first set of slave rollers or the second set of slave rollers based on selective engagement of at least one of the first independent pulley or the second independent pulley.

[0007] The present disclosure further provides a pulley for a motor driven roller (MDR) of a conveyor system, wherein the pulley comprising at least a fixed pulley comprising a first surface, a second surface, a first bearing hub, a rotating shaft, and a first bearing, wherein the first surface of the fixed pulley function as a first edge of the smart pulley; a first independent pulley comprising a third surface, a fourth surface, a first electromagnet clutch, a first electromagnet, a second bearing,

and a second bearing hub, wherein the third surface of the first independent pulley is coupled with the second surface of the fixed pulley; and a second independent pulley comprising a fifth surface, a sixth surface, a second electromagnet clutch, a second electromagnet, a third bearing, and a third bearing hub, wherein the fifth surface of the second independent pulley is coupled with the fourth surface of the first independent pulley and the sixth surface of the second independent pulley function as a second edge of the smart pulley.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The description of the illustrative embodiments can be read in conjunction with the accompanying figures. It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements are exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the figures presented herein, in which:

[0009] FIG. 1A illustrates an example conveyor system, in accordance with one or more embodiments described herein;

[0010] FIG. 1B illustrates an exemplary diagram that depicts conveyor system, in accordance with one or more embodiments described herein;

[0011] FIG. 2A to FIG. 2C illustrates various exemplary diagrams of a motor driven roller, in accordance with one or more embodiments described herein;

[0012] FIG. 2D illustrates an exemplary diagram that depicts a smart pulley for a motor driven roller, in accordance with one or more embodiments described herein;

[0013] FIG. 2E illustrates an exemplary diagram that depicts a sectional cut-out view of a smart pulley for a motor driven roller, in accordance with one or more embodiments described herein;

[0014] FIG. 2F illustrates an exemplary diagram that depicts another sectional cut-out view of a smart pulley for a motor driven roller, in accordance with one or more embodiments described herein;

[0015] FIG. 2G illustrates an exemplary diagram that depicts a cross-sectional view of a first part and a second part of a first independent pulley, in accordance with one or more embodiments described herein;

[0016] FIG. 3 illustrates a block diagram of an apparatus that includes a conveyer system, in accordance with one or more embodiments described herein;

[0017] FIG. 4A and FIG. 4B collectively illustrates various scenario-based diagrams for a conveyer system and a subsequent conveyer system, in accordance with one or more embodiments described herein;

[0018] FIG. 5A and FIG. 5B collectively illustrates scenario-based diagrams of a conveyer system and a subsequent conveyer system, in accordance with one or more embodiments described herein;

[0019] FIG. 6 illustrates a diagram of a conveyer system, in accordance with one or more embodiments described herein;

[0020] FIG. 7 illustrates a flow chart that depicts various operations of a conveyor system, in accordance with one or more embodiments described herein; and,

[0021] FIG. 8 illustrates a flow chart of a method of controlling a conveyor system, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION

[0022] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments.

However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments. The term “or” is used herein in both the alternative and conjunctive sense, unless otherwise indicated. The terms “illustrative,” “example,” and “exemplary” are used to be examples with no indication of quality level. Like numbers refer to like elements throughout.

[0023] The phrases “in an embodiment,” “in one embodiment,” “according to one embodiment,” and the like generally mean that the particular feature, structure, or characteristic following the phrase can be included in at least one embodiment of the present disclosure and can be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0024] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0025] If the specification states a component or feature “can,” “may,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that particular component or feature is not required to be included or to have the characteristic. Such component or feature can be optionally included in some embodiments, or it can be excluded.

[0026] In one or more embodiments, the present disclosure provides for an “Internet-of-Things” or “IoT” platform for industrial process control and/or industrial process optimization that uses real-time accurate models and/or real-time control for sustained peak performance of an enterprise and/or an enterprise process. The IoT platform is an extensible platform that is portable for deployment in any cloud or data center environment for providing an enterprise-wide, top to bottom control of processes and/or assets. Further, the IoT platform of the present disclosure supports end-to-end capability to execute models against process data and/or to translate the output into actionable insights and/or real-time control, as detailed in the following description.

[0027] A conveyor system is provided that includes a first set of slave rollers defining a first conveying section, a second set of slave rollers defining a second conveying section separate from the first conveying section. The conveyor system further includes a motor driven roller (MDR) that includes at least a first independent pulley, a second independent pulley, and a fixed pulley. Moreover, the first independent pulley is coupled with the first set of slave rollers and the second independent pulley is coupled with the second set of slave rollers. In addition, the motor driven roller is configured to drive at least one of the first set of slave rollers or the second set of slave rollers based on selective engagement of at least one of the first independent pulley or the second independent pulley. Beneficially as compared to conventional approaches, the conveyor system can efficiently drive the first set of slave rollers or the second set of slave rollers based on presence of the package, such as to improve the overall efficiency of the conveyor system with reduced cost and without depending on the length of the package. In addition, the conveyor system can be used to control different conveying sections with low initial cost and reduced maintenance cost.

[0028] FIG. 1A illustrates an example conveyor system **102**, according to one or more embodiments of the present disclosure. The conveyor system **102** may also be referred to as a roller conveyor system, a roller conveyor, and the like, which is configured to handle and move different type of packages, such as cardboard boxes, plastic crates, tires, and lightweight items from one place to another place. As shown in FIG. 1A, the conveyor system **102** includes a first set of slave rollers **104A**, **104B**, **104C**, **104D**, and **104E** (hereinafter referred to as the first set of slave rollers **104**), a second set of slave rollers **106A**, **106B**, **106C**, **106D**, and **106E** (hereinafter referred to as the second set of slave rollers **106**), a motor driven roller **108**, and a smart pulley **109**. Additionally, the conveyor system **102** is connected with a controller **110**, a server **114**, and a network **112**. In

some examples, the controller **110**, which is further connected to the server **114** through the network **112**. There is further shown that the smart pulley **109** of the motor driven roller **108** includes at least a first independent pulley **116A**, a second independent pulley **116B**, and a fixed pulley **118**. There is further shown a first edge **119A** and a second edge **119B** of the smart pulley **109**. There is further shown a belt **120**, an upstream photo eye sensor **121**, a first photo eye sensor **122A**, a second photo eye sensor **122B**, an upstream reflector **123**, a first reflector **124A**, a second reflector **124B**, a first frame **126A**, and a second frame **126B**. There is further shown an upstream conveyor system **128**, a package **130**, a first set of belts **132**, and a second set of belts **133**.

[0029] In an example embodiment, the first set of slave rollers **104** may define a first conveying section. Each slave roller from the first set of slave rollers **104** is conical or cylindrical shaped and is configured to handle different type of packages, such as cardboard boxes, plastic crates, tires, and lightweight items. Optionally, each slave roller from the first set of slave rollers **104** can be made of a metal, fiber, polymer, plastic, and the like. In one or more embodiments, each slave roller from the first set of slave rollers **104** includes a first end **134A** and a second end **134B**, such as the first end **134A** of each slave roller is arranged on the first frame **126A** of the conveyor system **102** and the second end **134B** of each slave roller is arranged on the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** are used to hold each slave roller from the first set of slave rollers **104**. In an example, length of the first conveying section along a first axis **138A** can vary from 24-inches to 30-inches. In another example, the length of the first conveying section along the first axis **138A** can vary from 30-inches to 36-inches. In yet another example, the length of the first conveying section along the first axis **138A** can vary from 36-inches to 48-inches.

[0030] In an example embodiment, the second set of slave rollers **106** may define a second conveying section separate from the first conveying section. In an implementation, each slave roller from the first set of slave rollers **104** as well as from the second set of slave rollers **106** may include pulley. In an example embodiment, each slave roller from the first set of slave rollers **104** as well as from the second set of slave rollers **106** may be referred to as slave rollers, which does not include any internal motor. Moreover, each slave roller from the first set of slave rollers **104** as well as from the second set of slave rollers **106** can be rotated in clockwise as well as anticlockwise direction. In one or more embodiments, each slave roller from the second set of slave rollers **106** includes a third end **136A** and a fourth end **136B**, such as the third end **136A** of each slave roller is arranged on the first frame **126A** of the conveyor system **102** and the fourth end **136B** of each slave roller is arranged on the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** are used to hold each slave roller from the second set of slave rollers **106**. In an example, a length of the second conveying section along a second x-axis **138B** can vary from 24-inches to 30-inches. In another example, the length of the second conveying section along the second x-axis **138B** can vary from 30-inches to 36-inches. In yet another example, the length of the second conveying section along the second x-axis **138B** can vary from 36-inches to 48-inches. Optionally, the length of the second conveying section along the second x-axis **138B** can be equal or different from the length of the first conveying section along the first x-axis **138A**.

[0031] In an example embodiment, the motor driven roller (MDR) **108** may be coupled to the first set of slave roller **104** and the second set of slave roller **106**. In some examples, the motor driven roller **108** may be configured to drive the first set of slave rollers **104** and the second set of slave roller **106**. In an example embodiment, the motor driven roller **108** is configured to drive at least one of the first set of slave rollers **104** or the second set of slave rollers **106** based on selective engagement of at least one of the first independent pulley **116A** or the second independent pulley **116B**, as further shown and described in FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6, and FIG. 7. Examples of implementations of each of the first independent pulley **116A** and the second independent pulley **116B** may include but are not limited to an 'o' shaped pulley, a 'v' shaped

pulley, a poly 'o' belt pulley, a poly 'v' belt pulley, or a timer belt, without limiting the scope of the present disclosure. In an example embodiment, the motor driven roller **108** may correspond to the shaft of a motor itself. In another embodiment, the motor driven roller **108** may correspond to a set of pulleys that may be driven by the motor, without departing from the scope of the disclosure. Accordingly, the motor driven roller **108** may correspond to any mechanism that is capable of driving the first set of slave roller **104** and the second set of slave rollers **106**. In one or more embodiments, the motor driven roller **108** includes a fifth end **140A** and a sixth end **140B**. In such embodiment, the fifth end **140A** of the motor driven roller **108** is coupled with the first frame **126A** of the conveyor system **102** and the sixth end **140B** of the motor driven roller **108** is coupled with the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** provides support to the motor driven roller **108**. Furthermore, in such embodiment, a belt **120** is coupled with the fixed pulley **118** of the motor driven roller **108**, such as the belt **120** is used to rotate the motor driven roller **108** through the fixed pulley **118**. In some examples, the scope of the disclosure is not limited to the motor driven roller **108** having one fixed pulley. In an alternative embodiment, the motor driven roller **108** may include more than one fixed pulley **118**, as is further described and depicted in FIG. **1B**. The structure of the motor driven roller **108** is further described in conjunction with FIG. **2A** to FIG. **2C**.

[0032] The controller **110** may be embodied as one or more microprocessors with accompanying digital signal processor(s), one or more processor(s) without an accompanying digital signal processor, one or more coprocessors, one or more multi-core processors, one or more controllers, processing circuitry, one or more computers, various other processing elements including integrated circuits such as, for example, an application specific integrated circuit (ASIC) or field programmable gate array (FPGA), or some combination thereof. Accordingly, in an example embodiment, the controller **110** may include a plurality of processors and signal processing modules. The plurality of processors may be embodied on a single electronic device or may be distributed across a plurality of electronic devices collectively configured to function as the circuitry of the conveyor system **102**. The plurality of processors may be in operative communication with each other and may be collectively configured to perform one or more functionalities of the circuitry of the conveyor system **102**, as described herein. In an example embodiment, the controller **110** may be configured to execute instructions stored in a memory or otherwise accessible to the controller **110**. These instructions, when executed by the controller **110**, may cause the circuitry of the conveyor system **102** to perform one or more of the functionalities, as described herein. Whether configured by hardware, firmware/software methods, or by a combination thereof, the controller **110** may include an entity capable of performing operations according to embodiments of the present disclosure while configured accordingly. Thus, for example, when the controller **110** is embodied as an ASIC, FPGA or the like, the controller **110** may include specifically configured hardware for conducting one or more operations described herein. Alternatively, as another example, when the controller **110** is embodied as an executor of instructions the instructions may specifically configure the controller **110** to perform one or more algorithms and operations described herein.

[0033] The controller **110** used herein may refer to a programmable microprocessor, microcomputer or multiple processor chip or chips that can be configured by software instructions (applications) to perform a variety of functions, including the functions of the various embodiments described above. In some devices, multiple processors may be provided dedicated to wireless communication functions and one processor dedicated to running other applications. Software applications may be stored in the internal memory before they are accessed and loaded into the processors. The processors may include internal memory sufficient to store the application software instructions. In many devices, the internal memory may be a volatile or nonvolatile memory, such as flash memory, or a mixture of both. The memory can also be located internal to another computing resource (e.g., enabling computer readable instructions to be downloaded over the

Internet or another wired or wireless connection). In one or more embodiments, the controller **110** is configured to determine the presence of the package **130** and control the conveyor system **102**. For example, the controller **110** is configured to determine the presence of the package **130** near the first set of slave rollers **104** and drive the motor driven roller **108** through the fixed pulley **118**. Furthermore, the controller **110** is configured to determine the presence of the package **130** on the first set of slave rollers **104** and drive the second set of slave rollers **106**. An exemplary implementation of the working of the controller **110** is further shown and described in FIG. 7.

[0034] In an embodiment, the network **112** may include a communication medium through which the conveyor system **102** may communicate with the server **114**. Such a communication may be performed, in accordance with various wired and wireless communication protocols. Examples of such wired and wireless communication protocols include, but are not limited to, Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), ZigBee, EDGE, infrared (IR), IEEE 802.11, 802.16, 2G, 3G, 4G, 5G, 6G cellular communication protocols, and/or Bluetooth (BT) communication protocols. The network **112** may include, but is not limited to, the Internet, a cloud network, a Wireless Fidelity (Wi-Fi) network, a Wireless Local Area Network (WLAN), a Local Area Network (LAN), a telephone line (POTS), and/or a Metropolitan Area Network (MAN).

[0035] In various embodiments, the server **114** can include any type or quantity of one or more processors and one or more data storage devices comprising memory for storing and executing applications or software modules to execute the operation of the conveyor system **102**. In one embodiment, the one or more processors and one or more data storage devices are embodied in the server **114**, such as enterprise-level servers. For example, in an embodiment, the processors and data storage devices comprise any type or combination of application servers, communication servers, web servers, super-computing servers, database servers, file servers, mail servers, proxy servers, and/virtual servers. Further, the one or more processors are configured to access the memory and execute processor-readable instructions, which when received and executed by the controller **110** to perform a plurality of functions of the conveyor system **102**.

[0036] In one or more embodiments, the upstream photo eye sensor **121** is arranged on the upstream conveyor system **128**, which is connected before the conveyor system **102**, as shown in FIG. 1A. Moreover, the upstream conveyor system **128** is further connected with the controller **110**. Examples of the upstream photo eye sensor **121** may include but are not limited to a photo sensor, a light sensor, a photodiode, a photoresistor, a phototransistor, a photovoltaic light sensor, and the like. The upstream photo eye sensor **121** is configured to determine the presence of the package **130** on the upstream conveyor system **128** and in a proximity region of the first set of slave rollers **104**. Optionally, the proximity region can vary from 5 centimeter (cm) to 10 cm, from 10 cm to 15 cm, or 15 cm to 20, and the like without limiting the scope of the present disclosure. In an example, the upstream photo eye sensor **121** works along with the upstream reflector **123** to detect the presence of the package **130** on the upstream conveyor system **128** and near (e.g., 10 cm to 15 cm away from) the first set of slave rollers **104** of the conveyor system **102**. In an example, if a photodiode is used as the upstream photo eye sensor **121**, then the upstream photo eye sensor **121** is configured to operate in visible and near infrared spectrum with a wavelength range from 400 nanometers (nm) to 1100 nm. In an example, the upstream photo eye sensor **121** is configured to work at a first wavelength that ranges from 401 nm to 410 nm. However, the first wavelength of the upstream photo eye sensor **121** may vary depending on a type of sensor being used. Furthermore, the upstream photo eye sensor **121** may be configured to determine if the package **130** is present in proximity region of the upstream photo eye sensor **121**, such as 10 cm to 15 cm away from the first set of slave rollers **104** of the conveyor system **102**. Thereafter, the upstream photo eye sensor **121** is configured to transmit the upstream photo eye signal to the controller **110**. An exemplary implementation of working of the upstream photo eye sensor **121** is further shown and described in FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6, and FIG. 7.

[0037] The first photo eye sensor **122A** is configured to determine the presence of the package **130** on the first set of slave rollers **104**. In an implementation, the first photo eye sensor **122A** is configured to work along with the first reflector **124A**, such as the first photo eye sensor **122A** is arranged on the first frame **126A** and the first reflector **124A** is arranged on the second frame **126B**. Examples of the first photo eye sensor **122A** may include but are not limited to a photo sensor, a light sensor, a photodiode, a photoresistor, a phototransistor, a photovoltaic light sensor, and the like. In one or more embodiments, the first photo eye sensor **122A** is arranged in proximity region of the first set of slave rollers **104** of the conveyor system **102**. Optionally, the proximity region can vary from 5 centimeter (cm) to 10 cm, from 10 cm to 15 cm, or 15 cm to 20 cm, and the like without limiting the scope of the present disclosure. In addition, each of the first photo eye sensor **122A** as well as the first reflector **124A** can be arranged on other positions without limiting the scope of the present disclosure. In an implementation, if a photodiode is used as the first photo eye sensor **122A**, then the first photo eye sensor **122A** is configured to operate in visible and near infrared spectrum with a wavelength range from 400 nm to 1100 nm. However, the wavelength of the first photo eye sensor **122A** may vary depending on a type of sensor being used. In an example, the first photo eye sensor **122A** is configured to work at a second wavelength that ranges from 411 nm to 420 nm, which is different from the first wavelength of the upstream photo eye sensor **121** (e.g., ranges from 401 nm to 410 nm). In addition, the first photo eye sensor **122A** along with the first reflector **124A** are used to detect the presence of the package **130** on the first set of slave rollers **104**. Furthermore, if the package **130** is present on the first set of slave rollers **104** of the conveyor system **102**, then the first photo eye sensor **122A** is configured to transmit the first photo eye signal to the controller **110**. Herein, the first photo eye signal transmitted by the first photo eye sensor **122A** is different from the upstream photo eye signal transmitted by the upstream photo eye sensor **121**.

[0038] In one or more embodiments, the second photo eye sensor **122B** is configured to determine the presence of the package **130** on the second set of slave rollers **106**. In an implementation, if a photodiode is used as the second photo eye sensor **122B**, then the second photo eye sensor **122B** is configured to operate in visible and near infrared spectrum with a wavelength between approximately 400 nm and 1100 nm. However, the wavelength of the second photo eye sensor **122B** may vary depending on a type of sensor being used. In an example, the second photo eye sensor **122B** is configured to work at a third wavelength that ranges can ranges from 421 nm to 430 nm, which is different from the first wavelength (e.g., ranges from 411 nm to 420 nm) of the upstream photo eye sensor and also different from the second wavelength (e.g., ranges from 401 nm to 410 nm) of the first photo eye sensor **122A**. In an implementation, the second photo eye sensor **122B** is configured to work with the second reflector **124B**, such as the second photo eye sensor **122B** is arranged on the first frame **126A** and the second reflector **124B** is arranged on the second frame **126B**. However, each of the second photo eye sensor **122B** as well as the second reflector **124B** can be arranged on other positions without limiting the scope of the present disclosure. In addition, the second photo eye sensor **122B** along with the second reflector **124B** are used to detect the presence of the package **130** on the second set of slave rollers **106**. For example, if the package **130** is present on the second set of slave rollers **106** of the conveyor system **102**, then the second photo eye sensor **122B** is configured to transmit the second photo eye signal to the controller **110**. Herein, the second photo eye signal transmitted by the second photo eye sensor **122B** is different from the upstream photo eye signal as well as from the first photo eye signal. In addition, in response to the second photo eye signal, the controller **110** is configured to continue rotation of the motor driven roller **108** through the fixed pulley **118** and also configured to drive the second set of slave rollers **106** based on selective engagement of the second independent pulley **108B**. An exemplary implementation of operation the conveyor system **102** is further shown and described in FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6, and FIG. 7.

[0039] FIG. 1B illustrates an exemplary diagram **100B** that depicts the conveyor system **102**, in

accordance with one or more embodiments of the present disclosure. As shown in FIG. 1B, the conveyor system **102** includes the first set of slave rollers **104**, the second set of slave rollers **106**, and the motor driven roller **108**. The conveyor system **102** as shown in FIG. 1B is same as that shown in FIG. 1A, such as the conveyor system **102** of FIG. 1B includes all components as previously shown and described in FIG. 1A. In addition to that, the conveyor system **102** of FIG. 1B further includes a first fixed pulley **118A** and a second fixed pulley **118B**.

[0040] With reference to FIG. 1B there is shown that the first independent pulley **116A** is arranged and located in proximity region of (e.g., 2 cm to 5 cm away from) the second frame **126B**, while the second independent pulley **116B** is arranged and located in proximity region of (e.g., 2 cm to 5 cm away from) the first frame **126A**. In addition, the motor driven roller **108** includes two different fixed pulleys, such as the first fixed pulley **118A** that is coupled with the first independent pulley **116A** and the second fixed pulley **118B** that is coupled with the second independent pulley **116B**, as shown in FIG. 1B. An exemplary implementation of the motor driven roller **108** with two different fixed pulleys, such as the first fixed pulley **118A** and the second fixed pulley **118B** is further shown and described in FIG. 2B. Furthermore, each of the first fixed pulley **118A** and the second fixed pulley **118B** can be same as that of the first fixed pulley **118A** of FIG. 1A.

[0041] In operation, the controller **110** (of FIG. 1A) is configured to determine the presence of the package **130** on the downstream conveyor system **128** and in proximity of the first set of slave rollers **104** (as previously described in FIG. 1A), such as based on a downstream photo eye signal, which is received from the upstream photo eye sensor **121**. In an implementation, in response to the downstream photo eye signal, the controller **110** may be configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, such as to rotate the motor driven roller **108**. In other words, in response to the downstream photo eye signal, the controller **110** is configured to rotate the motor driven roller **108** through the first fixed pulley **118A**, which is coupled with a belt **120A**. Thereafter, the controller **110** is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the second logical signal is different from the first logical signal, such as the second logical signal is transmitted to control the motor driven roller **108** to selectively engage the first independent pulley **116A** with the first fixed pulley **118A** of the motor driven roller **108**. Furthermore, as the motor driven roller **108** is already rotating, therefore, due to the rotation of the motor driven roller **108**, the first independent pulley **116A** also starts rotating. In addition, as the first independent pulley **116A** is coupled with the first set of slave rollers **104** through the first set of belts **132**, therefore due to the rotation of the first independent pulley **116A**, each slave roller in the first set of slave rollers **104** also starts rotating, for example through the first set of belts **132**. As a result, the motor driven roller **108** is configured to rotate each slave roller from the first set of slave rollers **104** to move the package **130** from the first set of slave rollers **104** towards the second set of slave rollers **106** through the motor driven roller **108**.

[0042] Thereafter, the controller **110** is configured to determine the presence of the package **130** on the first set of slave rollers **104**, for example based on a first photo eye signal, which is received from the first photo eye sensor **122** (as previously described in FIG. 1A). In an implementation, in response to the first photo eye signal, the controller **110** may be configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the third logical signal is different from the first logical signal and also different from the second logical signal. Moreover, the third logical signal is transmitted to control the motor driven roller **108** to selectively engage the second independent pulley **116B** with the second fixed pulley **118B** of the motor driven roller **108**. Furthermore, as the motor driven roller **108** is already rotating, therefore, due to the rotation of the motor driven roller **108**, the second independent pulley **116B** also starts rotating. In addition, as the second independent pulley **116B** is already coupled with the second set of slave rollers **106** through the second set of belts **133**, therefore due to the rotation of the second independent pulley **116B**, each slave roller in the second set of slave rollers **106** also starts rotating,

for example through the second set of belts **133**. As a result, the package **130** can be shifted from the first set of slave rollers **104** on the second set of slave rollers **106** through the motor driven roller **108**.

[0043] FIGS. 2A to 2C illustrate various exemplary diagrams of the motor driven roller **108**, in accordance with one or more embodiments of the present disclosure. With reference to FIG. 2A, there is shown a diagram **200A** that depicts the motor driven roller **108**, which includes a body **201**, an inbuilt motor, such as a motor **202**, an axle **204**, and the smart pulley **109**. The smart pulley **109** (or a pulley) further include the first edge **119A** and the second edge **119B**, which are perpendicular to a third x-axis **203** and aligned towards a y-axis **205**. Herein, the second edge **119B** of the smart pulley **109** is arranged at the fifth end **140A** of the motor driven roller **108**. There is further shown that the smart pulley **109** further includes the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118**.

[0044] The motor driven roller **108** of FIG. 2A is a cylindrical shaped roller, which includes the body **201**, which is aligned towards the third x-axis **203**. In an implementation, the motor driven roller **108** may also be referred to as a roller. Optionally, the body **201** of the motor driven roller **108** can be made of a metal, fiber, polymer, or any other material without limiting the scope of the present disclosure. There is further shown that the axle **204** is inserted within the body **201** of the motor driven roller **108**, such as inserted from the fifth end **140A** of the motor driven roller **108** and aligned towards the third x-axis **203**, as shown in FIG. 2A. As a result, the axle **204** is centrally aligned within the body **201** and also centrally aligned with the fifth end **140A** of the motor driven roller **108**. Furthermore, the axles **204** passes from the first edge **119A** of the smart pulley **109** towards the second edge **119B** of the smart pulley **109**. There is further shown that, the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118** of the smart pulley **109** are also coupled around the axle **204**, which is further shown and described in FIG. 2D to FIG. 2F. There is further shown that the motor **202** is arranged within the body **201** of the motor driven roller **108** and aligned towards the third x-axis **203**. Moreover, the motor **202** can be coupled with the axle **204** of the motor driven roller **108**. There is further shown that the motor **202** is arranged near the sixth end **140B** of the motor driven roller **108**, for example, the motor **202** can be arranged around 2 cm to 4 cm away from sixth end **140B** of the motor driven roller **108**. Optionally the motor **202** can be arranged around 4 cm to 6 cm (or around 4 cm to 8 cm) away from sixth end **140B** of the motor driven roller **108**, without limiting the scope of the present disclosure.

[0045] The smart pulley **109** includes the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118**. In an example, the smart pulley **109** may also be referred to as a pulley, a gear, and the like. Similarly, each of the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118** may also be referred to as a drum, a gear, a pulley, and the like. In an implementation, the smart pulley **109** is externally coupled with the motor **202**, as further shown and described in FIG. 5B. Furthermore, there is shown in FIG. 2A that the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118** of the smart pulley **109** are arranged in proximity of the fifth end **140A** of the motor driven roller **108**. An exemplary implementation of arrangement of each of the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118** is further shown and described in FIG. 2D to 2F. In an implementation, the motor driven roller **108** can include two or more smart pulleys without limiting the scope of the present disclosure.

[0046] In an implementation, the motor **202** is a high torque electric motor, which is used to drive (i.e., rotate) the motor driven roller **108** through the smart pulley **109**, such as based on the one or more instructions received from the controller **110**, as further shown and described in FIG. 4A and FIG. 4B. Examples of implementation of the motor **108** may include but are not limited to a stepper motor, a direct current motor, an alternate current motor, a servo motor, a direct drive motor, a linear motor, and the like. In an implementation, the motor driven roller **108** can include two or

more motors without limiting the scope of the present disclosure. In addition, the axle **204** may be configured to drive each of the first independent pulley **116A**, the second independent pulley **116B**, the fixed pulley **118** through the motor **202**. Examples of the axle **204** may include but are not limited to a rod, a pipe, a pole, a shaft, and the like.

[0047] With reference to FIG. 2B there is shown that the first independent pulley **116A** is arranged adjacent to the fifth end **140A** of the motor driven roller **108**, while the second independent pulley **116B** is arranged adjacent to the sixth end **140B** (i.e., opposite to the fifth end **140A**) of the motor driven roller **108**. In addition, with reference to FIG. 2B, there is further shown that the motor driven roller **108** includes two fixed pulleys, such as the first fixed pulley **118A** and the second fixed pulley **118B**. Moreover, the first fixed pulley **118A** is coupled with the first independent pulley **116A** and is arranged adjacent to the fifth end **140A** of the motor driven roller **108**.

Similarly, the second fixed pulley **118B** is coupled with the second independent pulley **116B** and is arranged adjacent to the sixth end **140B** of the motor driven roller **108**. In addition, with reference to FIG. 2C, there is shown that, the motor driven roller **108** does not include any inbuilt motor, therefore an external motor may be required to drive the motor driven roller **108** of FIG. 2C.

Furthermore, an axle **206** of the exemplary diagram **200C** is configured to drive each of the first independent pulley **116A**, the second independent pulley **116B**, the fixed pulley **118** through the external motor, which is further shown and described in FIGS. 4A and 4B. In an example, axle **206** may be same as that of the axle **204** of FIG. 2A and FIG. 2B. Optionally, the motor driven roller **108** of the exemplary diagram **200C** may be referred to as a slave roller, a roller, and the like.

[0048] FIG. 2D illustrates an exemplary diagram **200D** that depicts the smart pulley **109** for the motor driven roller **108**, in accordance with one or more embodiments of the present disclosure. FIG. 2E illustrates an exemplary diagram **200E** that depicts a sectional cut-out view of the smart pulley **109** for the motor driven roller **108**, in accordance with one or more embodiments of the present disclosure.

[0049] With reference to FIG. 2D, there is shown that the smart pulley **109** includes the fixed pulley **118**, which includes the first independent pulley **116A**, the second independent pulley **116B**, and the fixed pulley **118**. There is further shown that the fixed pulley **118** include a first surface **208A** and a second surface **208B**. There is further shown that the first independent pulley **116A** include a third surface **209A** and a fourth surface **209B**. Similarly, the second independent pulley **116B** include a fifth surface **210A** and a sixth surface **210B**. Moreover, each of the first surface **208A**, the second surface **208B**, the third surface **209A**, the fourth surface **209B**, the fifth surface **210A** and a sixth surface **210B** are perpendicular to the third x-axis **203**, as shown in FIG. 2D.

[0050] With reference to FIG. 2D, there is shown that the fixed pulley **118** is coupled around the axle **204**, such as the first surface **208A** of the fixed pulley **118** is arranged and located at one end **207A** of the axle **204**. In addition, the first surface **208A** of the fixed pulley **118** may function as the first edge **119A** (of FIG. 1A) of the smart pulley **109**. In addition, the first independent pulley **116A** is also coupled with the fixed pulley **118** and coupled around the axle **204**. For example, the third surface **209A** of the first independent pulley **116A** is coupled with the second surface **208B** of the fixed pulley **118**, as shown in FIG. 2D. There is further shown that the second independent pulley **116B** is coupled with the first independent pulley **116A** and also coupled around at an outermost end **207B** of the axle **204**. For example, the fifth surface **210A** of the second independent pulley **116B** is coupled with the fourth surface **209B** of the first independent pulley **116A**. Moreover, the sixth surface **210B** of the second independent pulley **116B** may function as the second edge **119B** (of FIG. 1A) of the smart pulley **109**, which is arranged and located at the outermost end **207B** of the axle **204**. Therefore, the first independent pulley **116A** is arranged between the fixed pulley **118** and the second independent pulley **116B**, as shown in FIG. 2D.

[0051] With reference to FIG. 2E, there is shown the body **201**, the axle **204**, a cable **211**, a first bearing hub **212**, a rotating shaft **214**, and a first bearing **216**. There is further shown a first electromagnet clutch **220**, a second bearing **222**, a first electromagnet **224**, and a second bearing

hub **226**. With reference to FIG. 2E, there is further shown, a y-axis **227**, a second electromagnet clutch **228**, a third bearing **230**, a second electromagnet **232**, and a third bearing hub **234**.

[0052] In one or more embodiments, if the axle **204** is a hollow axle, then the cable **211** passes through the axle **204**. In an example, if the axle **204** is a rod-shaped axle, then the cable **211** can be arranged outside the motor driven roller **108**. Optionally, the cable **211** may also be referred to as a wire, a fiber, and may be made of copper, aluminum, or any other conducting medium. In an example, the cable **211** is used to supply power from an external power source to the motor **202** of the motor driven roller **108**, such as to drive the motor **202** of the motor driven roller **108**.

[0053] In one or more embodiments, the fixed pulley **118** of the motor driven roller **108** includes the first bearing hub **212**, the rotating shaft **214**, and the first bearing **216**. In an embodiment, the first bearing hub **212** is an outer portion of the first bearing **216**. Furthermore, the first bearing hub **212** is a rotatable section, which is configured to rotate the motor driven roller **108**. Moreover, the first bearing hub **212** is coupled with the body **201** of the motor driven roller **108**.

[0054] In one or more embodiments, the rotating shaft **214** may be made from metal, polymer, fiber, or any other material. In an example, the rotating shaft **214** is a part of the fixed pulley **118**. In another example, the rotating shaft **214** can be externally coupled with the fixed pulley **118**. In an implementation, the rotating shaft **214** is configured to rotate the fixed pulley **118** of the motor driven roller **108**.

[0055] In one or more embodiments, the first bearing **216** may also be referred to as angular contact bearing, a load bearing, and the like. In an example, diameter of the first bearing **216** along the y-axis **227** can vary from 30 millimeters (mm) to 35 mm. However, the diameter of the first bearing **216** can have any other value without limiting the scope of the present disclosure. There is further shown that the first bearing **216** includes one or more rotatable balls **218**, which are arranged within the first bearing **216** and are configured to rotate the second bearing **222**.

Moreover, the first bearing **216** and the rotating shaft **214** are coupled with the first bearing hub **212**. Therefore, each of the first bearing hub **212**, the rotating shaft **214**, and the first bearing **216** are collectively configured to rotate the motor driven roller **108** through the fixed pulley **118**.

[0056] In one or more embodiments, the first electromagnet clutch **220** is coupled with the rotating shaft **214** of the motor driven roller **108**. Examples of the first electromagnet clutch **220** may include but are not limited to an electromagnetic clutch, a bearing clutch, a fluid clutch, a friction clutch, a single plate clutch, a multiplate clutch, and the like. In an implementation, the first electromagnet clutch **220** is inactive in the absence of electro-magnetic field and is configured to get activated in the presence of electro-magnetic field.

[0057] In one or more embodiments, the second bearing **222** may be similar or different from the first bearing **216**. In an example, diameter of the second bearing **222** along the y-axis **227** can vary from 40 mm to 50 mm, which is higher than the diameter (i.e., 30 mm to 35 mm) of the first bearing **216**. However, the diameter of the second bearing **222** along the y-axis **227** can have any other value without limiting the scope of the present disclosure. In an implementation, the second bearing **222** is arranged in proximity of the first electromagnet clutch **220**, for example 1 mm to 2 mm away from the first electromagnet clutch **220**, as shown in FIG. 2E. There is further shown that the second bearing **222** includes one or more rotatable balls **223**, which are arranged within the second bearing **222** and are configured to rotate the second bearing **222**.

[0058] In one or more embodiments, the first electromagnet **224** is referred to as a magnet that includes coils, such as when electric current passing through the coils, then the coils start producing magnetic field. Furthermore, in normal operation, the first electromagnet **224** is not coupled with the first electromagnet clutch **220**. However, the first electromagnet **224** is coupled with the second bearing hub **226**. In an example, the second bearing hub **226** is made of polymer and may also be referred to as a poly-v bearing hub assembly, a poly-o bearing hub assembly, and the like.

Moreover, structure and properties of the second bearing hub **226** may be same as that of the first bearing hub **212**. In an embodiment, the second bearing hub **226** is an outer portion of the second

bearing **222**. Furthermore, the second bearing hub **226** is a rotatable section, which is configured to rotate the second independent pulley **116B**.

[0059] In one or more embodiments, the second electromagnet clutch **228** of the second independent pulley **116B** may be same as that of the first electromagnet clutch **220** of the first independent pulley **116A**. However, the structure of the second electromagnet clutch **228** can be different from the structure of the first electromagnet clutch **220** without limiting the scope of the present disclosure. In addition, diameter of the third bearing **230** along the y-axis **227** can vary from 30 millimeter (mm) to 35 mm, which is smaller than the diameter of the second bearing **222**. However, the diameter of the third bearing **230** can have any other value without limiting the scope of the present disclosure. There is further shown that the third bearing **230** includes one or more rotatable balls **218**, which are arranged within the third bearing **230** and are configured to rotate the third bearing **230**. Furthermore, structure and properties of the second electromagnet **232** are also similar as that of the first electromagnet **224**. In addition, the second electromagnet **232** is attached on the third bearing hub **234** of the second independent pulley **116B**. Furthermore, in normal operation, the second electromagnet clutch **228** is coupled with the rotating shaft **214** of the motor driven roller **108**, while not coupled with the second electromagnet **232**.

[0060] In one or more embodiments, the upstream photo eye sensor **121** is configured to determine the presence of the package **130** on the upstream conveyor system **128** and in a proximity region of the first set of slave rollers **104**, as previously described in FIG. 1A. Thereafter, the upstream photo eye sensor **121** is configured to transmit the upstream photo eye signal to the controller **110**. In an implementation, in response to the downstream photo eye signal, the controller **110** is configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, due to which the rotating shaft **214** starts rotating. In addition, the rotating shaft **214** is configured to drive the motor driven roller **108**, such as through the first bearing **216** and the first bearing hub **212**, which is further connected to the belt **120A**. Thereafter, the controller **110** is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, as previously described in FIG. 1B. In an implementation, the second logical signal is transmitted to control the motor driven roller **108**, such as to activate the first electromagnet **224** of the first independent pulley **116A**. As a result, the first electromagnet **224** gets activated and starts generating electro-magnetic field. Furthermore, due to effect of the as generated electro-magnetic field, the first electromagnet clutch **220** gets engaged with the first electromagnet **224**. As a result, the first independent pulley **116A** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the first independent pulley **116A** also starts rotating, such as through the first electromagnet clutch **220** and the first electromagnet **224**. In an implementation, the first electromagnet clutch **220** is configured to rotate the first independent pulley **116A** through the second bearing **222**. Furthermore, the first independent pulley **116A** is coupled with the first set of slave rollers **104** (of FIG. 1A), for example through the first set of belts **132**, which are coupled around the second bearing hub **226**. Therefore, due to the rotation of the first independent pulley **116A**, each slave roller in the first set of slave rollers **104** also starts rotating, for example through the first set of belts **132**. As a result, the first independent pulley **116A** along with the motor driven roller **108** is configured to rotate each slave roller from the first set of slave rollers **104** to move the package **130** from the first set of slave rollers **104** towards the second set of slave rollers **106** through the motor driven roller **108**.

[0061] In one or more embodiments, the controller **110** is further configured to determine the presence of the package **130** on the first set of slave rollers **104**, for example based on a first photo eye signal, which is received from the first photo eye sensor **122** (as previously described in FIGS. 1A and 1B). In an implementation, in response to the first photo eye signal, the controller **110** may be configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the third logical signal is different from the first logical signal as well as from the second logical signal. In an implementation, the third logical signal is transmitted to

control the motor driven roller **108**, such as to activate the second electromagnet **232**. As a result, the second electromagnet **232** gets activated and starts generating electro-magnetic field. Furthermore, due to the effect of the as generated electro-magnetic field, the second electromagnet clutch **228** gets engaged with the second electromagnet **232**. Therefore, the second independent pulley **116B** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the second independent pulley **116B** also starts rotating, such as through the second electromagnet clutch **228** and the second electromagnet **232**. In an implementation, the second electromagnet clutch **228** may be configured to rotate the second independent pulley **116B** through the third bearing **230**. Furthermore, the second independent pulley **116B** is coupled with the second set of slave rollers **106**, for example through the second set of belts **133**. Therefore, due to the rotation of the second independent pulley **116B**, each slave roller in the second set of slave rollers **106** also starts rotating, for example through the second set of belts **133**. As a result, the second independent pulley **116B** along with the motor driven roller **108** is configured to rotate each slave roller from the second set of slave rollers **106** through the third bearing hub **234**.

[0062] In one or more embodiments, if the package **130** is not present on the first set of slave rollers **104** (of FIG. **1A**), then the first photo eye sensor **122** may not transmit the first photo eye signal. Therefore, in response to the absence of the package **130** on the first set of slave rollers **104**, the controller **110** will not receive the first photo eye signal and configured to transmit a fourth logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the fourth logical signal is different from each of the first logical signal, the second logical signal, and the third logical signal. Furthermore, the fourth logical signal is transmitted to control the motor driven roller **108**, such as to deactivate the first electromagnet **224** of the first independent pulley **116A**. Therefore, the first electromagnet **224** will stop generating the electro-magnetic field. Furthermore, due to the absence of the electro-magnetic field, the first electromagnet **224** gets disengaged from the first electromagnet clutch **220**. As a result, the motor driven roller **108** is configured to disengage the first independent pulley **116A** from the rotating shaft **214** of the fixed pulley **118**, due to which the first electromagnet clutch **220** stops rotating the first independent pulley **116A**. Therefore, due to disengagement from the rotating shaft **214**, the first independent pulley **116A** stop rotating and each slave roller from the first set of slave rollers **104** also stop rotating. Beneficially, as compared to conventional approaches, the first independent pulley **116A** can be used to precisely control and precisely drive each slave roller from the first set of slave rollers **104**.

[0063] FIG. **2F** illustrates an exemplary diagram **200F** that depicts another sectional cut-out view of the smart pulley **109** for the motor driven roller **108**, in accordance with one or more embodiments of the present disclosure. As shown in FIG. **2F**, the motor driven roller **108** includes the first independent pulley **116A**, the second independent pulley **116B**, the fixed pulley **118**, the axle **204**, and the cable **211**. There is further shown that the first independent pulley **116A** include a first part **236**, a second part **238**, a first set of springs **240**, a first set of electromagnets **242**. In addition, the first part **236** includes a first set of protrusions **237** while the second part **238** includes a first set of grooves **239**, which is further shown and described in FIG. **2G**. Furthermore, the second independent pulley **116B** include a third part **244**, a fourth part **246**, a second set of springs **248**, and a second set of electromagnets **250**. Moreover, the third part **244** include a second set of protrusions **245** while the fourth part **246** include a second set of grooves **247**, which is further shown and described in FIG. **2G**.

[0064] In one or more embodiments, the first independent pulley **116A** includes the first part **236A** and the second part **238**, such as the first part **236** and the second part **238** are facing each other, as shown in FIG. **2F**. In addition, the first part **236** is coupled with the rotating shaft **214** of the fixed pulley **118** while the second part **238** is coupled with the second bearing hub **226**. In an example, each of the first part **236A** and the second part **238** may be made from metal, polymer, fiber, and the like. In an implementation, the second part **238** of the first independent pulley **116A** includes

the first set of electromagnets **242**. In an example, the first set of electromagnets **242** can have similar properties as that of the first electromagnet **224** and the second electromagnet **232** of FIG. 2E. Furthermore, the first independent pulley **116A** may also include the first set of springs **240** which are configured to be connected between the first part **236** and the second part **238**. In an example, the first set of springs **240** are flexible in nature and can be made from flexible metal, polymer, and the like. In addition, in normal operation, the first set of springs **240** are in expanded position, due to which the second part **238** is not engaged with the first part **236**, as shown in FIG. 2F. As a result, in normal operation, the first independent pulley **116A** remains in steady state and does not rotate.

[0065] In one or more embodiments, the second independent pulley **116B** includes the third part **244** and the fourth part **246**. Furthermore, the third part **244** is coupled with the rotating shaft **214** and the fourth part **246** is coupled with the fourth bearing hub **234**, as shown in FIG. 2F. Optionally, the third part **244** can have similar properties as that of the first part **236** while the fourth part **246** can have similar properties as that of the second part **238** respectively. In an implementation, the fourth part **246** includes the second set of electromagnets **250**, which can have similar properties as that of the first electromagnet **224** and the second electromagnet **232**. Furthermore, the second independent pulley **116B** may also include the second set of spring **248**, which have similar properties as that of the first set of springs **240**, such as the second set of spring **248** are coupled between the third part **244** and the fourth part **246**. In addition, in normal operation, the second set of spring **248** are in expanded position, due to which the third part **244** is not engaged with the fourth part **246**. As a result, in normal operation, the second independent pulley **116B** remains in steady state and does not rotate.

[0066] In one or more embodiments, the upstream photo eye sensor **121** is configured to determine the presence of the package **130** on the upstream conveyor system **128** and in a proximity region of the first set of slave rollers **104**, as previously described in FIG. 1A. Thereafter, the upstream photo eye sensor **121** is configured to transmit the upstream photo eye signal to the controller **110**. In an implementation, in response to the downstream photo eye signal, the controller **110** is configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, due to which the rotating shaft **214** starts rotating, as previously described in FIG. 2F. In addition, the rotating shaft **214** is configured to drive the motor driven roller **108**, such as through the first bearing **216** and the first bearing hub **212**, which is further connected to the belt **120A**. Thereafter, the controller **110** is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, as previously described in FIG. 1B. In an implementation, the second logical signal is transmitted to control the motor driven roller **108**, such as to activate the first set of electromagnets **242** of the first independent pulley **116A**. As a result, the first set of electromagnets **242** starts generating electro-magnetic field. In an implementation, once the first set of electromagnets **242** starts generating electro-magnetic field, then the first set of protrusions **237** of the first part **236** gets engaged within the first set of grooves **239** of the second part **238**. In an implementation, in the presence of the as generated electro-magnetic field, the first set of springs **240** also gets compressed, due to which the first set of protrusions **237** of the first part **236** gets engaged within the first set of grooves **239** of the second part **238**. Furthermore, as the first part **236** is already coupled with the rotating shaft **126**, therefore, due to the engagement of the first part **236** with the second part **238**, the first independent pulley **116A** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. As a result, the first independent pulley **116A** starts rotating, which further drive each slave roller from the first set of slave rollers **104** (of FIG. 1A).

[0067] In one or more embodiments, the controller **110** is configured to determine the presence of the package **130** on the first set of slave rollers **104**, for example based on a first photo eye signal, which is received from the first photo eye sensor **122** (as previously described in FIGS. 1A and 1B). In an implementation, in response to the first photo eye signal, the controller **110** may be configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven

roller **108**. Herein the third logical signal is different from the first logical signal and different from the second logical signal. In an implementation, the third logical signal is transmitted to control the motor driven roller **108**, such as to activate the second set of electromagnets **250** of the second independent pulley **116B**, due to which the second set of electromagnets **250** starts generating electro-magnetic field. In an implementation, once the second set of electromagnets **250** starts generating electro-magnetic field, then the second set of protrusions **245** of the third part **244** gets engaged within the second set of grooves **247** of the fourth part **246**. In an implementation, in the presence of as generated electro-magnetic field, the second set of springs **248** also gets compressed, due to which the second set of protrusions **245** of the third part **244** gets engaged within the second set of grooves **247** of the fourth part **246**. Furthermore, as the third part **244** is already coupled with the rotating shaft **126**, therefore, due to the engagement of the third part **244** with the fourth part **246**, the second independent pulley **116B** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. As a result, the second independent pulley **116B** starts rotating, which further drive each slave roller from the second set of slave rollers **106** (of FIG. 1A).

[0068] FIG. 2G illustrates an exemplary diagram **200G** that depicts a cross-sectional view of the first part **236** and the second part **238** of the first independent pulley **116A**, in accordance with one or more embodiments of the present disclosure. As shown in FIG. 2G, the first part **236** includes the first set of protrusions **237** and the first part **236** includes the first set of grooves **239** as well as one or more stopper **252**, such as the one more stopper **252** are configured to stop the first set of protrusions **237** of the first part **236** within the first set of grooves **239** of the second part **238**. In an implementation, shape at an end side of each of the first set of protrusions **237** is conical shape, while shape at an end side of each of the one more stopper **252** is bevel shape. As a result, when the first set of protrusions **237** exactly within the first set of grooves **239** till the one more stopper **252** in receiver, then due to the conical shape at the end of each protrusion and each stopper, the first set of protrusions **237** starts rotating. There is further shown that the first part **236** is circular in shape, which includes a first side **254A** and a second side **254B**, such as the first set of protrusions **237** are grown on the first side **254A** of the first part **236**. There is further shown that the first set of protrusions **237** are equally spaced from the center of the first side **254A** of the first part **236**. There is further shown that the second part **238** is also circular in shape, which includes a third side **256A** and a fourth side **256B**, such as the first set of grooves **239** are created on the third side **256A** of the second part **238**. There is further shown that the first set of grooves **239** are also equally spaced from the center of the first side **254A** of the first part **236**. However, dimensions as well as arrangement of the first set of protrusions **237** as well as the first set of grooves **239** may vary without limiting the scope of the present disclosure. Herein, the exemplary diagram **200G** depicts the cross-sectional view of the first part **236** and the second part **238**. Furthermore, the cross-sectional view is same for the third part **244** and the fourth part **246** of FIG. 2F.

[0069] FIG. 3 illustrates a block diagram of an apparatus **300** that includes the conveyer system **102**, in accordance with one or more embodiments of present disclosure. FIG. 3 illustrates the block diagram that depicts an apparatus **300** that includes the conveyer system **102**. There is further shown that the apparatus **300** includes the controller **110**, a memory **302**, and an interface card **304**. [0070] The apparatus **300** can include a plurality of conveyer systems, such as the controller **110** is configured to control operation of each conveyer system based on one or more instructions received from the server **114** through the network **112**. Moreover, each conveyer system from the plurality of conveyer systems includes a motor drive roller similar as that of the motor driven roller **108**. Optionally, each conveyer system from the plurality of conveyer systems can work in a similar way as that of the conveyer system **102**.

[0071] The memory **302** is configured to store one or more instructions to control operation of the conveyer system **102**, and to control other conveyer systems. Optionally, the one more instruction can be embodied by one or more processor-executable software modules that reside on the server **114**. By way of example but not limitation, the memory **302** can include a random-access memory

(RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, disk storage, magnetic storage devices, or the like. Disk storage, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and the like or other storage devices that store data magnetically or optically with lasers. Combinations of the above types of media are also included within the scope of the terms non-transitory computer-readable and processor-readable media. Additionally, any combination of instructions stored on the one or more non-transitory processor-readable or computer-readable media can be referred to herein as a computer program product.

[0072] The interface card **304** may also be referred to as a network interface card, a communication interface, and the like. Moreover, the interface card **304** is configured to provide an interface between the conveyer system **102**, the controller **110**, and the memory **302**. In an implementation, the interface card **304** can also be used to provide communication between the apparatus **300** and the server **114** through the network **112** (of FIG. 1A). According to various embodiments, the interface card **304** can include one or more cellular radios, Bluetooth, WiFi, near-field communication radios, Ethernet, or other appropriate communication devices for transmitting and receiving information between two points of communication.

[0073] FIG. 4A and FIG. 4B collectively illustrates various scenario-based diagrams for the conveyer system **102** and a subsequent conveyer system **402**, in accordance with one or more embodiments of the present disclosure. FIG. 4A illustrates a diagram **400A** that depicts a front view of the conveyer system **102**, which is coupled with the subsequent conveyer system **402**. Moreover, FIG. 4B illustrates a diagram **400B** that depicts a side view of the conveyer system **102** that is connected with the subsequent conveyer system **402**. There is further shown that the subsequent conveyer system **402** includes a third set of slave rollers **404**, a fourth set of slave rollers **406**, and a subsequent motor driven roller **408**. With reference to FIG. 4B, there is further shown, a third photo eye sensor **410A**, a fourth photo eye sensor **410B**, a third reflector **412A**, a fourth reflector **412B**, a third independent pulley **414**, a fourth independent pulley **416**, a subsequent fixed pulley **418**, and a subsequent belt **420**. In addition, with reference to FIG. 4B, there is shown, an external motor **422**, a shaft **424**, a first set of pulleys **426**, and a second set of pulleys **428**.

[0074] The subsequent conveyer system **402** is same as that of the conveyer system **102**. Therefore, the third set of slave rollers **404** and the fourth set of slave rollers **406** also same as that of the first set of slave rollers **104** and the second set of slave rollers **106**. In addition, the subsequent motor driven roller **408** is same as that of the motor driven roller **108** of FIG. 1A to FIG. 2C. Similarly, the third photo eye sensor **410A** and the fourth photo eye sensor **410B** have similar properties as that of the first photo eye sensor **122A** and the second photo eye sensor **122B** respectively. Furthermore, the third reflector **412A** and the fourth reflector **412B** have similar properties as that of the first reflector **124A** and the second reflector **124B** respectively. In an implementation, each of the third photo eye sensor **410A** along with the third reflector **412A** are configured to determine the presence of the package **130** on the third set of slave rollers **404**. Similarly, the fourth photo eye sensor **410B** along with the fourth reflector **412B** are configured to determine the presence of the package **130** on the fourth set of slave rollers **406**. In addition, the third independent pulley **414** and the fourth independent pulley **416** have similar properties as that of the first independent pulley **116A** and the second independent pulley **116B**. Similarly, the subsequent fixed pulley **418** have similar properties as that of fixed pulley **118**. In addition, the subsequent belt **420** is also configured to work in a similar way as that of the belt **120**. In an embodiment, the belt **120** is coupled with the fixed pulley **118** of the motor driven roller **108** as well as with the shaft **424** of an external motor **422**, as shown in FIG. 4B. In an implementation, the belt **120** is used to couple the motor driven roller **108** with the external motor **422** and through the first set of pulleys **426**. In such implementation, the belt **120** is coupled with the motor driven roller **108** as well as with the shaft **424** of the external motor **422** through the first set of pulleys **426**. Therefore, the belt **120** is beneficial to drive (i.e., rotate) the motor driven roller **108** using the external motor **422** and

through the fixed pulley **118**. Examples of implementations of the external motor **422** may include, but are not limited to a stepper motor, a direct current motor, an alternate current motor, a servo motor, a direct drive motor, a linear motor, and the like. Herein, the external motor **422** is further connected with the controller **110** (of FIG. **1A**)

[0075] In operation, the controller **110** (of FIG. **1A**) is configured to receive an upstream photo eye signal from the upstream photo eye sensor **121** based on the presence of the package **130** near the first set of slave rollers **104**. In an implementation, in response to the downstream photo eye signal, the controller **110** is configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to drive the external motor **422**, such as to rotate the motor driven roller **108** through the first set of pulleys **426** that are coupled with the shaft **424** of the external motor **422**. Thereafter, the controller **110** is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. In an implementation, the second logical signal is transmitted to control the motor driven roller **108**, such as to activate the first electromagnet **224** of the first independent pulley **116A**. As a result, the first electromagnet **224** gets activated and starts generating electro-magnetic field. Furthermore, due to effect of as generated electro-magnetic field, the first electromagnet clutch **220** gets engaged with the first electromagnet **224**. As a result, the first independent pulley **116A** gets selectively engaged with the rotating shaft **214** of the fixed pulley **118**, as previously described in FIGS. **2F** and **2G**. Furthermore, due to the rotation of the rotating shaft **214**, the first independent pulley **116A** also starts rotating, such as through the first electromagnet clutch **220** and the first electromagnet **224**. As a result, the first independent pulley **116A** along with the motor driven roller **108** is configured to rotate each slave roller from the first set of slave rollers **104** to move the package **130** from the first set of slave rollers **104** towards the second set of slave rollers **106** through the motor driven roller **108**.

[0076] The controller **110** is further configured to receive a first photo eye signal from the first photo eye sensor **122A** based on the presence of the package **130** on the first set of slave rollers **104**. Thereafter, in response to the first photo eye signal, the controller **110** is configured to continue rotation of the motor driven roller **108** through the fixed pulley **118** as well as through the first set of pulleys **426** that are coupled with the external motor **422**, as shown in FIG. **4B**. In an implementation, in response to the first photo eye signal, the controller **110** may be configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, as previously described in FIGS. **2F** and **2G**. In an implementation, the third logical signal is transmitted to control the motor driven roller **108**, such as to activate the second electromagnet **232**. As a result, the second electromagnet **232** gets activated and starts generating electro-magnetic field. Furthermore, due to the effect of the as generated electro-magnetic field, the second electromagnet clutch **228** gets engaged with the second electromagnet **232**. Therefore, the second independent pulley **116B** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the second independent pulley **116B** also starts rotating, as previously described in FIG. **2F** and FIG. **2G**. In addition, the second independent pulley **116B** is already coupled with the second set of slave rollers **106**, for example through the second set of belts **133**. Therefore, due to the rotation of the second independent pulley **116B**, each slave roller in the second set of slave rollers **106** also starts rotating, for example through the second set of belts **133**. As a result, the second independent pulley **116B** along with the motor driven roller **108** is configured to drive each slave roller from the second set of slave rollers **106**, such as to move the package **130** from first set of slave rollers **104** towards the second set of slave rollers **106** and also from the second set of slave rollers **106** towards the subsequent conveyor system **402**. Thereafter, the controller **110** is further configured to receive a second photo eye signal from the second photo eye sensor **122B** based on the presence of the package **130** on the second set of slave rollers **106**. Furthermore, in response to the second photo eye signal, the controller **110** is further configured to determine, through the first photo eye sensor **122A**, if the package **130** is present on the first set of slave rollers **104**. In addition, in response to presence of the package **130**,

the controller **110** is further configured to control the motor driven roller **108** to continue engagement of the first independent pulley **116A** and continue rotation of each slave roller from the first set of slave rollers **104**.

[0077] In one or more embodiments, if the package **130** is not present on the first set of slave rollers **104** (of FIG. 1A), then the first photo eye sensor **122** may not transmit the first photo eye signal. Therefore, in response to the absence of the package **130** on the first set of slave rollers **104**, the controller **110** may be configured to transmit a fourth logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the fourth logical signal is different from each of the first logical signal, the second logical signal, and the third logical signal. Furthermore, the fourth logical signal is transmitted to control the motor driven roller **108**, such as to deactivate the first electromagnet **224** of the first independent pulley **116A**. Therefore, the first electromagnet **224** will stop generating the electro-magnetic field. Furthermore, due to the absence of the electro-magnetic field, the first electromagnet **224** gets disengaged from the first electromagnet clutch **220**. As a result, the motor driven roller **108** is configured to disengage the first independent pulley **116A** from the rotating shaft **214** of the fixed pulley **118**, due to which the first electromagnet clutch **220** stops rotating the first independent pulley **116A**. Therefore, due to disengagement from the rotating shaft **214**, the first independent pulley **116A** stop rotating and each slave roller from the first set of slave rollers **104** also stop rotating. Beneficially as compared to conventional approaches, the first independent pulley **116A** can be used to precisely control and precisely drive each slave roller from the first set of slave rollers **104**. However, due to the rotation of the second independent pulley **116B**, the package **130** can be shifted from the second set of slave rollers **106** towards the subsequent conveyor system **402**, while each slave roller from the first set of slave rollers **104** remains in steady state and does not rotate.

[0078] In response to the second photo eye signal from the second photo eye sensor **122B** based on the presence of the package **130** on the second set of slave rollers **106**, the controller **110** is configured to rotate the subsequent motor driven roller **408** through the subsequent fixed pulley **418**. In an example, the subsequent motor driven roller **408** is rotated through the second set of pulleys **428** that are coupled with the shaft **424** of the motor **422** through the subsequent belt **420**, as shown in FIG. 4B. In addition, in response to the second photo eye signal, the controller **110** is configured to control the subsequent motor driven roller **408** to selectively engage the third independent pulley **414** (similarly as that of the first independent pulley **116A**). Furthermore, based on the selective engagement of the third independent pulley **514**, the controller **110** is configured to rotate each slave roller from the third set of slave rollers **404**, such as to move the package **130** from the second set of slave rollers **106** towards the third set of slave rollers **404**. Thereafter, the controller **110** is further configured to receive a third photo eye signal from the third photo eye sensor **410A**, which works along with the third reflector **412A**. Furthermore, in response to the third photo eye signal, the controller **110** is configured to determine that the package **130** is present on the third set of slave rollers **404**. As a result, the controller **110** is configured to continue rotation of each slave roller from the third set of slave rollers **404**, such as to move the package **130** from the third set of slave rollers **404** to the fourth set of slave rollers **406**. In addition, the controller **110** is configured to control the subsequent motor driven roller **408** to selectively engage the fourth independent pulley **416** (similarly as that of the second independent pulley **116B**). Thereafter, (similarly as that of the fourth independent pulley **416**, the controller **110** is configured to rotate each slave roller from the fourth set of slave rollers **406**, such as to move the package **130** from the third set of slave rollers **404** towards the fourth set of slave rollers **406**. In addition, due to the presence of the package **130** on the third set of slave rollers **404**, the controller **110** is configured to continue rotation of each slave roller from the third set of slave rollers **404**.

[0079] The controller **110** is further configured to receive a fourth photo eye signal from the fourth photo eye sensor **410B**, which works along with the fourth reflector **412B**. Thereafter, in response to the fourth photo eye signal, the controller **110** is configured to determine that the package **130** is

present on the fourth set of slave rollers **406**. As a result, the controller **110** is configured to continue rotation of each slave roller from the fourth set of slave rollers **406**. However, due to the absence of any package on the third set of slave rollers **404**, the controller **110** may not receive the third photo eye signal. As a result, the controller **110** is configured to control the subsequent motor driven roller **408** to disengage the third independent pulley **414** and stop rotation of each slave roller from the third set of slave rollers **404**.

[0080] FIG. 5A and FIG. 5B collectively illustrates scenario-based diagrams of the conveyer system **102** and the subsequent conveyer system **402**, according to the present disclosure. FIG. 5A illustrates a diagram **500A** that depicts a front view of the conveyer system **102**, which is coupled with the subsequent conveyer system **402**. Moreover, FIG. 5B illustrates a diagram **500B** that depicts a side view of the conveyer system **102** that is coupled with the subsequent conveyer system **402**. With reference to FIG. 5A and FIG. 5B there is shown a third roller **502A**, a fourth roller **502B**, a fifth roller **504A**, a sixth roller **504B**, a first belt **506A**, a second belt **506B**, a third belt **508A**, a fourth belt **508B**, a subsequent smart pulley **510**, a fifth belt **512A**, and a sixth belt **512B**.

[0081] The conveyer system **102** includes the third roller **502A**, which is further connected to the first set of slave rollers **104**. The conveyer system **102** further includes the fourth roller **502B**, which is further connected to the second set of slave rollers **106**, as shown in FIGS. 5A and 5B. Moreover, each of the third roller **502A** and the fourth roller **502B** may corresponds to the motor driven roller **108** of FIG. 2A to FIG. 2C. Optionally, each of the third roller **502A** and the fourth roller **502B** may corresponds to a slave roller from the first set of slave rollers **104** or from the second set of slave rollers **106** (i.e., a roller that does not include the smart pulley **109**). In addition, the third roller **502A** as well as the fourth roller **502B** are connected to the shaft **424** of the external motor **422** through the smart pulley **109** of FIG. 2A to FIG. 2F. Herein, the smart pulley **109** is not coupled directly with the third roller **502A** as well as with the fourth roller **502B**, but the smart pulley **109** is arranged separately from the third roller **502A** as well as from the fourth roller **502B**, as shown in FIG. 5B. There is further shown that the third roller **502A** is coupled with the first independent pulley **116A** of the smart pulley **109** through the first belt **506A** while the fourth roller **502B** is coupled with the second independent pulley **116B** of the smart pulley **109** through the second belt **506B**. Furthermore, the fixed pulley **118** of the smart pulley **109** is connected to the external motor **422** through the fifth belt **512A**, as shown in FIG. 5B.

[0082] Similarly, the subsequent conveyer system **402** includes the fifth roller **504A**, which is further coupled to the third set of slave rollers **404**. The subsequent conveyer system **402** further includes the sixth roller **504B**, which is further coupled to the fourth set of slave rollers **406**, as shown in FIG. 5A and FIG. 5B. Moreover, each of the fifth roller **504A** and the sixth roller **504B** may corresponds to the motor driven roller **108** of FIG. 2A to FIG. 2C. Optionally, each of the fifth roller **504A** and the sixth roller **504B** may corresponds to a slave roller from the first set of slave rollers **104** or from the second set of slave rollers **106** (i.e., a roller that does not include the smart pulley **109**). Moreover, the fifth roller **504A** as well as the sixth roller **504B** are coupled to the shaft **424** of the external motor **422** through the subsequent smart pulley **510**, as shown in FIG. 5B. In an example, the subsequent smart pulley **510** includes all the components as that of the smart pulley **109**. Moreover, the fifth roller **504A** is coupled to the subsequent smart pulley **510** through the third belt **508A** while the sixth roller **504B** is coupled to the subsequent smart pulley **510** through the fourth belt **508B**. Furthermore, the subsequent smart pulley **510** is also coupled to the external motor **422** through the sixth belt **512B**, as shown in FIG. 5B.

[0083] In operation, the controller **110** (of FIG. 1A) is configured to receive an upstream photo eye signal from the upstream photo eye sensor **121** (of FIG. 1A) based on the presence of the package **130** in proximity of the first set of slave rollers **104**, as previously explained in FIG. 4A and FIG. 4B. In an implementation, in response to the downstream photo eye signal, the controller **110** is configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to drive the external

motor **422**, such as to rotate the smart pulley **109**, for example through the fifth belt **512A**. Thereafter, in response to the upstream photo eye signal, the controller **110** is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the smart pulley **109**, such as to selectively engage the first independent pulley **116A** of the smart pulley **109** with the fixed pulley **118**, due to which the first independent pulley **116A** starts rotating, as previously explained in FIG. 2F and FIG. 2G. Furthermore, rotation of the first independent pulley **116A** of the smart pulley **109** results in rotation of the first belt **506A**, due to which the third roller **502A** starts rotating. Furthermore, due to the rotation of the third roller **502A**, each slave roller in the first set of slave rollers **104** also starts rotating. As a result, the package **130** is shifted towards the first set of slave rollers **104**. Thereafter, the controller **110** is further configured to receive a first photo eye signal from the first photo eye sensor **122A** based on the presence of the package **130** on the first set of slave rollers **104**, as previously described in FIG. 4A and FIG. 4B. Furthermore, in response to the first photo eye signal, the controller **110** is configured to continue rotation of the third roller **502A**, due to which each slave roller in the first set of slave rollers **104** also keeps rotating. Thereafter, in response to the first photo eye signal, the controller **110** is configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the smart pulley **109**, such as to selectively engage the second independent pulley **116B** of the smart pulley **109** with the fixed pulley **118**, due to which the fourth roller **502B** starts rotating through the second belt **506B**. Furthermore, due to the rotation of the fourth roller **502B**, each slave roller in the second set of slave rollers **106** also starts rotating. As a result, the package **130** is shifted from the first set of slave rollers **104** of the first conveying section towards the second set of slave rollers **106** in the second conveying section. [0084] The controller **110** is configured to receive a second photo eye signal from the second photo eye sensor **122B** based on the presence of the package **130** on the second set of slave rollers **106**. Furthermore, in response to the second photo eye signal, the controller **110** is further configured to control the subsequent smart pulley **510**, such as to rotate the fifth roller **504A** through the third belt **508A**. Herein, operation of the subsequent smart pulley **510** is controlled by the controller **110** in a similar way as that of controlling the operation of the smart pulley **109**. Furthermore, due to the rotation of the fifth roller **504A**, each slave roller in the third set of slave rollers **404** also starts rotating. As a result, the package **130** is shifted from the second set of slave rollers **106** in second conveying section towards the third set of slave rollers **404** in the third conveying section. Therefore, the controller **110** is further configured to receive a third photo eye signal from the third photo eye sensor **410A** based on the presence of the package **130** on the third set of slave rollers **404**. In addition, in response to the third photo eye signal, the controller **110** is further configured to selectively engage the subsequent smart pulley **510** that results in rotation of the fourth belt **508B**, due to which the sixth roller **504B** starts rotating. Furthermore, due to the rotation of the sixth roller **504B**, each slave roller in the fourth set of slave rollers **406** also starts rotating. As a result, the package **130** is shifted from the third set of slave rollers **404** in third conveying section towards the fourth set of slave rollers **406** in the fourth conveying section.

[0085] FIG. 6 illustrates a diagram **600** of a conveyer system **602**, in accordance with one or more embodiments of present disclosure. FIG. 6 illustrates a diagram **600** that depicts a front view as well as a side view of the conveyer system **602**. There is further shown a first conveyer belt **604A**, a second conveyer belt **604B**, a third conveyer belt **604C**, a first gap filler **606A**, and a second gap filler **606B**.

[0086] The first conveyer belt **604A** is configured to collectively couple each slave roller in the first set of slave rollers **104**, the second conveyer belt **604B** is configured to collectively couple each slave roller in the second set of slave rollers **106**, and the third conveyer belt **604C** is configured to collectively couple each slave roller in the third set of slave rollers **404**. With reference to FIG. 6, there is further shown a side view, which depicts that the motor driven roller **108** and the subsequent motor driven roller **408** are not aligned with the first set of slave rollers **104**, the second set of slave rollers **106**, and the third set of slave rollers **404**. Due to which, there exists a first gap

between the height of the motor driven roller **108** and the first set of slave rollers **104** and also with the height of the second set of slave rollers **106**. Similarly, there exists a second gap between the height of the subsequent motor driven roller **408** and the second set of slave rollers **106** and also with the height of the third set of slave rollers **404**. In an example, the first gap filler **606A** is arranged in the first gap and the second gap filler **606B** is arranged in the second gap, such as to fill the corresponding gaps.

[0087] In operation, the controller **110** (of FIG. 1A) is configured to rotate the motor driven roller **108** through the fixed pulley **118** as well as control the motor driven roller **108** to selectively engage the first independent pulley **116A**, due to which the first independent pulley **116A** starts rotating. Furthermore, rotation of the first independent pulley **116A** of the smart pulley **109** results in rotation of the first belt **506A**, due to which each slave roller in the first set of slave rollers **104** also starts rotating, such as through the first conveyer belt **604A**. Thereafter, the controller **110** is further configured to receive a first photo eye signal, such as to continue rotation of the first independent pulley **116A** of the smart pulley **109**, due to which each slave roller in the first set of slave rollers **104** also keeps rotating, for example due to the rotation of the first belt **506A** and the rotation of the first conveyer belt **604A**. Thereafter, in response to the first photo eye signal, the controller **110** is configured to control the smart pulley **109** to selectively engage the second independent pulley **116B** of the smart pulley **109**, due to which each slave roller in the second set of slave rollers **106** also starts rotating through the second conveyer belt **604B**. As a result, the package **130** is shifted from the first set of slave rollers **104** of the first conveying section towards the second set of slave rollers **106** in the second conveying section.

[0088] The controller **110** is further configured to receive a second photo eye signal based on the presence of the package **130** in the second conveying section. Furthermore, in response to the second photo eye signal, the controller **110** is further configured to control the subsequent smart pulley **510** to rotate the subsequent motor driven roller **408** and control the subsequent motor driven roller **408** to selectively engage the subsequent smart pulley **510**, due to which each slave roller in the third set of slave rollers **404** also starts rotating through the third conveyer belt **604C**. As a result, the package **130** is shifted from the second set of slave rollers **106** towards the third set of slave rollers **404**.

[0089] FIG. 7 illustrates a flow chart **700** that depicts various operations of the conveyor system **102**, in accordance with one or more embodiments of the present disclosure. As shown in FIG. 7, the flow chart **700** includes steps **702-to-720**. In an implementation, the controller **110** of the conveyor system **102** is configured to execute the steps **702-to-720**.

[0090] In one or more embodiments, if the package **130** is present on the upstream photo eye sensor **121** and in proximity region the first set of slave rollers **104** of the conveyor system **102**, then at step **702**, the upstream photo eye sensor **121** does not receive a reflected signal of the first wavelength from the upstream reflector **123**. Optionally, the proximity region can vary from 5 centimeter (cm) to 10 cm, from 10 cm to 15 cm, or 15 cm to 20, and the like without limiting the scope of the present disclosure. As a result, the upstream photo eye sensor **121** is configured to generate an upstream photo eye signal and transmit the upstream photo eye signal to the controller **110**. In an implementation, the upstream photo eye sensor **121** is configured to receive the reflected signal of the first wavelength continuously from the upstream reflector **123**, which depicts that the package **130** is not present in the proximity region of the upstream photo eye sensor **121**. As a result, if the upstream photo eye sensor **121** does not detect the presence of the package **130**, then the step **702** is not executed. Furthermore, if the package **130** is present on the first set of slave rollers **104**, then the first photo eye sensor **122A** will not receive a reflected signal of the second wavelength from the first reflector **124A**. Thereafter, at step **704**, a first photo eye signal is generated by the first photo eye sensor **122A**, which is further configured to transmit the first photo eye signal to the controller **110**. In addition, if the package **130** is present on the second set of slave rollers **106**, then the second photo eye sensor **122B** will not receive a reflected signal of the third

wavelength from the second reflector **124B**. As a result, at step **706**, a second photo eye signal is generated by the second photo eye sensor **122B**, which is further configured to transmit second photo eye signal to the controller **110**.

[0091] In an implementation, if the package **130** is present in proximity (e.g., 10 cm to 15 cm away from) the first set of slave rollers **104**, then at step **708**, the controller **110** is configured to receive the upstream photo eye signal (as generated at step **702**) from the upstream photo eye sensor **121**. However, if the package **130** is present on the first set of slave rollers **104**, then at step **708**, the controller **110** is configured to receive the first photo eye signal (as generated at step **704**) from the first photo eye sensor **122A**. Furthermore, if the package **130** is present on the second set of slave rollers **104**, then at step **708**, the controller **110** is further configured to receive the second photo eye signal (as generated at step **706**) from the second photo eye sensor **122B**.

[0092] Furthermore, at step **710**, the controller **110** is configured to determine if any of the upstream photo eye signal, the first photo eye signal, or the second photo eye signal is received or not. In addition, at step **710**, the controller **110** is further configured to determine, whether to drive the motor driven roller **108** or not, such as based on any of the upstream photo eye signal, the first photo eye signal or the second photo eye signal. In an implementation, at step **710**, if no package is detected in any of the first set of slave rollers **104** or the second set of slave rollers **106**, then the controller **110** does not receive any of the upstream photo eye signal, the first photo eye signal, or the second photo eye signal. Therefore, at step **712** is executed, where the controller **110** is configured to not to rotate the motor driven roller **108**. However, if at step **710**, the package **130** (or any other package) is detected on any of the first set of slave rollers **104** or the second set of slave rollers **106**, then the controller **110** receives any of the upstream photo eye signal, the first photo eye signal, or the second photo eye signal. Thereafter, at step **714**, the controller **110** is configured to determine, which independent pulley need to be driven. In an example, if at step **710**, the package **130** is present in the first set of slave rollers **104**, then at step **716**, the controller is configured to transmit a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, as previously explained in FIG. 2F and FIG. 2G. Herein the second logical signal is transmitted to activate the first electromagnet **224** of the first independent pulley **116A**. As a result, the first electromagnet **224** gets activated and starts generating electro-magnetic field. Furthermore, due to effect of as generated electro-magnetic field, the first electromagnet clutch **220** gets engaged with the first electromagnet **224**. As a result, the first independent pulley **116A** gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the first independent pulley **116A** also starts rotating, such as through the first electromagnet clutch **220** and the first electromagnet **224**. In an implementation, the first electromagnet clutch **220** is configured to rotate the first independent pulley **116A** through the second bearing **222**.

Furthermore, the first independent pulley **116A** is already coupled with the first set of slave rollers **104** (of FIG. 1A), for example through the first set of belts **132**, which are coupled around the second bearing hub **226**. Therefore, due to the rotation of the first independent pulley **116A**, each slave roller in the first set of slave rollers **104** also starts rotating, for example through the first set of belts **132**. In other words, the motor driven roller **108** is configured to rotate each slave roller from the first set of slave rollers **104** to move the package **130** from the first set of slave rollers **104** towards the second set of slave rollers **106** through the motor driven roller **108**.

[0093] In an implementation, if at step **710**, the package **130** is present in the second set of slave rollers **106**, then at step **718**, the controller **110** is configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, such as to activate the second electromagnet **232**. As a result, the second electromagnet **232** gets activated and starts generating electro-magnetic field. Furthermore, due to the effect of as the generated electro-magnetic field, the second electromagnet clutch **228** gets engaged with the second electromagnet **232**. Therefore, the second independent pulley **116B** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the second independent pulley **116B**

also starts rotating, such as through the second electromagnet clutch **228** and the second electromagnet **232**. In an implementation, the second electromagnet clutch **228** may be configured to rotate the second independent pulley **116B** through the third bearing **230**. Furthermore, the second independent pulley **116B** is already coupled with the second set of slave rollers **106**, for example through the second set of belts **133**. Therefore, due to the rotation of the second independent pulley **116B**, each slave roller in the second set of slave rollers **106** also starts rotating, for example through the second set of belts **133**. Therefore, the second independent pulley **116B** is configured to rotate each slave roller from the second set of slave rollers **106** through the third bearing hub **234**.

[0094] In an implementation, if the length along the first axis **138A** of the first conveying section that includes the first set of slave rollers **104** is 24-inches, the length along the first axis **138A** of the second conveying section that includes the second set of slave rollers **106** is also 24-inches, and width of the motor driven roller **108** along the y-axis **205** is 4-inches. Therefore, the overall length of the conveyor system **102** will be 52 inches. Furthermore, if the length of the package **130** is more than 24-inches (or more than 28 inches), then the package **130** will occupy a large space on the first set of slave rollers **104** as well as on the second set of slave rollers **106**. Therefore, at step **720**, the controller **110** is configured transmit the first logical signal as well as the second signal to the motor driven roller **108** to selectively engage each of the first independent pulley **116A** as well as the second independent pulley **116B**. In an implementation, if one package is present in the first conveying section and another package is present in the second conveying section, even then at step **720**, the controller **110** is further configured transmit the first logical signal as well as the second signal to the motor driven roller **108** to selectively engage each of the first independent pulley **116A** as well as the second independent pulley **116B**.

[0095] FIG. **8** illustrates a flow chart for a method **800** of controlling the conveyor system **102**, according to the present disclosure. As shown in FIG. **8**, the flow chart **800** includes steps **802**-to-**920**. In an implementation, the controller **110** of the conveyor system **102** is configured to execute the steps **802**-to-**814**.

[0096] At step **802**, the method **800** comprises, defining a first conveying section by the first set of slave rollers **104**. In an implementation, each slave roller from the first set of slave rollers **104** is conical or cylindrical shaped, which are configured to handle different type of packages, such as cardboard boxes, plastic crates, tires, and lightweight items. Optionally, each slave roller from the first set of slave rollers **104** can be made of a metal, fiber, polymer, plastic, and the like. In one or more embodiments, each slave roller from the first set of slave rollers **104** includes the first end **134A** and the second end **134B**, such as the first end **134A** of each slave roller is arranged on the first frame **126A** of the conveyor system **102** and the second end **134B** of each slave roller is arranged on the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** are used to support each slave roller from the first set of slave rollers **104**. In an example, length of the first conveying section along a first axis **138A** can vary from 24-inches to 30-inches. In another example, the length of the first conveying section along the first axis **138A** can vary from 30-inches to 36-inches. In yet another example, the length of the first conveying section along the first axis **138A** can vary from 36-inches to 48-inches.

[0097] At step **804**, the method **800** comprises, defining a second conveying section by the second set of slave rollers **106**, wherein the second conveying section is defined separately from the first conveying section. In an example embodiment, each slave roller from the first set of slave rollers **104** as well as from the second set of slave rollers **106** may be referred to as slave rollers, which does not include any internal motor. Moreover, each slave roller from the first set of slave rollers **104** as well as from the second set of slave rollers **106** can be rotated in clockwise as well as anticlockwise direction. In one or more embodiments, each slave roller from the second set of slave rollers **106** includes the third end **136A** and the fourth end **136B**, such as the third end **136A** of each slave roller is arranged on the first frame **126A** of the conveyor system **102** and the fourth end

136B of each slave roller is arranged on the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** can be used to hold each slave roller from the second set of slave rollers **106**. In an example, a length of the second conveying section along the second x-axis **138B** can vary from 24-inches to 30-inches. In another example, the length of the second conveying section along the second x-axis **138B** can vary from 30-inches to 36-inches. In yet another example, the length of the second conveying section along the second x-axis **138B** can vary from 36-inches to 48-inches. Optionally, the length of the second conveying section along the second x-axis **138B** can be equal or different from the length of the first conveying section along the first x-axis **138A**.

[0098] At step **806**, the method **800** comprises, arranging the motor driven roller **108** between the first set of slave rollers **104** and the second set of slave roller **106**, such as the motor driven roller **108** includes at least the fixed pulley **118**, the first independent pulley **116A**, and the second independent pulley **116B**. In an example embodiment, the motor driven roller **108** may be coupled to the first set of slave roller **104** and the second set of slave roller **106**. In some examples, the motor driven roller **108** may be configured to drive the first set of slave rollers **104** and the second set of slave roller **106**. In an example embodiment, the motor driven roller **108** is configured to drive at least one of the first set of slave rollers **104** or the second set of slave rollers **106** based on selective engagement of at least one of the first independent pulley **116A** or the second independent pulley **116B**. Examples of implementations of each of the first independent pulley **116A** and the second independent pulley **116B** may include but are not limited to a 'o' shaped pulley, a 'v' shaped pulley, a poly 'o' belt pulley, a poly 'v' belt pulley, or a timer belt, without limiting the scope of the present disclosure. In an example embodiment, the motor driven roller **108** may correspond to the shaft of a motor itself. In another embodiment, the motor driven roller **108** may correspond to a set of pulleys that may be driven by the motor, without departing from the scope of the disclosure. Accordingly, the motor driven roller **108** may correspond to any mechanism that is capable of driving the first set of slave roller **104** and the second set of slave rollers **106**. In one or more embodiments, the motor driven roller **108** includes the fifth end **140A** and the sixth end **140B**. In such embodiment, the fifth end **140A** of the motor driven roller **108** is coupled with the first frame **126A** of the conveyor system **102** and the sixth end **140B** of the motor driven roller **108** is coupled with the second frame **126B** of the conveyor system **102**. Therefore, each of the first frame **126A** as well as the second frame **126B** provides support to the motor driven roller **108**. Furthermore, in such embodiment, the belt **120** is coupled with the fixed pulley **118** of the motor driven roller **108**, such as the belt **120** is used to rotate the motor driven roller **108** through the fixed pulley **118**. In some examples, the scope of the disclosure is not limited to the motor driven roller **108** having one fixed pulley. In an alternative embodiment, the motor driven roller **108** may include more than one fixed pulley **118**.

[0099] At step **808** the method **800** comprises, coupling the first independent pulley **116A** of the motor driven roller **108** with the first set of slave rollers **104**. At step **810**, the method **800** comprises, coupling the second independent pulley **116B** of the motor driven roller **108** with the second set of slave rollers **106**. In accordance with an embodiment, the method **800** comprises, coupling the first independent pulley **116A** with each slave roller from the first set of slave rollers **104** through the first set of belts **132** and coupling the second independent pulley **116B** with each slave roller from the second set of slave rollers **106** through the second set of belts **133**. In an implementation, the first independent pulley **116A** is coupled with the first set of slave rollers **104** through the first set of belts **132**. In an implementation, the second independent pulley **116B** is coupled with the second set of slave rollers **106** through the second set of belts **133**.

[0100] In one or more embodiments, the method **800** comprises, receiving an upstream photo eye signal from the upstream photo eye sensor **121** (of FIG. **1A**) based on the presence of the package **130** in proximity of the first set of slave rollers **104** and in response to the upstream photo eye signal, driving the motor driven roller **108** through the fixed pulley **118**. In an implementation, the

upstream photo eye sensor **121** (of FIG. **1A**) is configured to determine the presence of the package **130** on the upstream conveyor system **128** and in a proximity region of the first set of slave rollers **104**, as previously described in FIG. **1A**. Thereafter, the upstream photo eye sensor **121** is configured to transmit the upstream photo eye signal to the controller **110**. In an implementation, in response to the downstream photo eye signal, the controller **110** is configured to transmit a first logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**, due to which, the rotating shaft **214** starts rotating. In addition, the rotating shaft **214** is configured to drive the motor driven roller **108**, such as through the first bearing **216** and the first bearing hub **212**, which is further connected to the belt **120A**. In accordance with an embodiment, the method **800** further comprises, driving the motor driven roller **108** through the inbuilt motor **202** of the motor driven roller **108** and through the fixed pulley **118**. In accordance with an embodiment, the method **800** further comprises, driving the motor driven roller **108** through the external motor **422** and through the fixed pulley **118**, such as the external motor **422**, which is connected with the motor driven roller **108**.

[0101] At step **812**, the method **800** comprises, selectively engaging the first independent pulley **116A** with the fixed pulley **118** of the motor driven roller **108** and driving, through the motor driven roller **108**, the first set of slave rollers **104** based on selective engagement of the first independent pulley **116A**. In accordance with an embodiment, the method **800** further comprises, controlling the motor driven roller **108** for activating the first electromagnet **224** of the first independent pulley **116A**. The first electromagnet **224** is activated for selectively engaging the first independent pulley **116A** with the rotating shaft **214** of the motor driven roller **108** through the first electromagnet clutch **220** and for driving each slave roller from the first set of slave rollers **104** through the second bearing **222**, the second bearing hub **226**, and the first set of belts **132**. In an implementation, the method **800** comprises, transmitting, by the controller **110** a second logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. In an implementation, the second logical signal is transmitted for controlling the motor driven roller **108**, such as for activating the first electromagnet **224** of the first independent pulley **116A**. As a result, the first electromagnet **224** gets activated and starts generating electro-magnetic field. In an implementation, due to effect of as generated electro-magnetic field, the first electromagnet clutch **220** gets engaged with the first electromagnet **224**. As a result, the first independent pulley **116A** gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the first independent pulley **116A** also starts rotating, such as through the first electromagnet clutch **220** and the first electromagnet **224**. As a result, the motor driven roller **108** is configured to rotate each slave roller from the first set of slave rollers **104** to move the package **130** from the first set of slave rollers **104** towards the second set of slave rollers **106** through the motor driven roller **108**.

[0102] At step **814**, the method **800** comprises, selectively engaging the second independent pulley **116B** with the fixed pulley **118** of the motor driven roller **108** and driving, through the motor driven roller **108**, the second set of slave rollers **106** based on selective engagement of the second independent pulley **116B**. In accordance with an embodiment, the method **800** further comprises, receiving a first photo eye signal from the first photo eye sensor **122A** based on the presence of the package **130** on the first set of slave rollers **104**. Moreover, in response to the first photo eye signal, controlling the motor driven roller **108** for activating the second electromagnet **232** of the second independent pulley **116B**. The second electromagnet **232** is activated for selectively engaging the second independent pulley **116B** with the rotating shaft **214** of the motor driven roller **108** through the second electromagnet clutch **228** and for driving each slave roller from the second set of slave rollers **106** through the third bearing **230**, the third bearing hub **234**, and the second set of belts **133**. In an implementation, the controller **110** is configured to receive a first photo eye signal from the first photo eye sensor **122A** based on the presence of the package **130** on the first set of slave rollers **104**. Thereafter, in response to the first photo eye signal, the controller **110** is configured to

continue rotation of the motor driven roller **108** through the fixed pulley **118** as well as through the first set of pulleys **426** that are coupled with the external motor **422**. In an implementation, in response to the first photo eye signal, the controller **110** may be configured to transmit a third logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. In an implementation, the third logical signal is transmitted to control the motor driven roller **108**, such as to activate the second electromagnet **232**. As a result, the second electromagnet **232** gets activated and starts generating electro-magnetic field. Furthermore, due to the effect of as the generated electro-magnetic field, the second electromagnet clutch **228** gets engaged with the second electromagnet **232**. Therefore, the second independent pulley **116B** also gets engaged with the rotating shaft **214** of the fixed pulley **118**. Furthermore, due to the rotation of the rotating shaft **214**, the second independent pulley **116B** also starts rotating, as previously described in FIG. 2F and FIG. 2G. Furthermore, the second independent pulley **116B** is already coupled with the second set of slave rollers **106**, for example through the second set of belts **133**. Therefore, due to the rotation of the second independent pulley **116B**, each slave roller in the second set of slave rollers **106** also starts rotating, for example through the second set of belts **133**. As a result, the motor driven roller **108** is configured to drive each slave roller from the second set of slave rollers **106**, such as to move the package **130** from first set of slave rollers **104** towards the second set of slave rollers **106** and also from the second set of slave rollers **106** towards the subsequent conveyor system **402**.

[0103] In accordance with an embodiment, the method **800** further comprises, receiving a second photo eye signal from the second photo eye sensor **122B** based on the presence of the package **130** on the second set of slave rollers **106**. In accordance with an embodiment, the method **800** further comprises, determining, through the first photo eye sensor **122A**, if the package **130** is present on the first set of slave rollers **104** and in response to presence of the package **130**. In addition, the method **800** comprises, controlling the motor driven roller **108** to continue engagement of the first independent pulley **116A** with the rotating shaft **214** of the motor driven roller **108** to drive each slave roller from the first set of slave rollers **104**. In accordance with an embodiment, the method **800** further comprises, determining, through the first photo eye sensor **122A**, if the package **130** is not present on the first set of slave rollers **104**. Moreover, in response to absence of the package **130**, controlling the motor driven roller **108** for deactivating the first electromagnet **224** of the first independent pulley **116A**. The first electromagnet **224** is deactivated for disengaging the first independent pulley **116A** from the rotating shaft **214** of the motor driven roller to stop driving each slave roller from the first set of slave rollers **104**. In an implementation, in response to the absence of the package **130** on the first set of slave rollers **104**, the controller **110** may be configured to transmit a fourth logical signal (e.g., either logic '1' or logic '0') to the motor driven roller **108**. Herein the fourth logical signal is different from each of the first logical signal, the second logical signal, and the third logical signal. Furthermore, the fourth logical signal is transmitted to control the motor driven roller **108**, such as to deactivate the first electromagnet **224** of the first independent pulley **116A**. Therefore, the first electromagnet **224** will stop generating the electro-magnetic field. Furthermore, due to the absence of the electro-magnetic field, the first electromagnet **224** gets disengaged from the first electromagnet clutch **220**. As a result, the motor driven roller **108** is configured to disengage the first independent pulley **116A** from the rotating shaft **214** of the fixed pulley **118**, due to which the first electromagnet clutch **220** stops rotating the first independent pulley **116A**. Therefore, due to disengagement from the rotating shaft **214**, the first independent pulley **116A** stop rotating, due to which each slave roller from the first set of slave rollers **104** also stop rotating. Therefore, due to the absence of the electro-magnetic field, the first independent pulley **116A** also gets disengaged (i.e., not connected) and stop rotation of each slave roller from the first set of slave rollers **104**. Beneficially as compared to conventional approaches, the first independent pulley **116A** can be used to precisely control and precisely drive each slave roller from the first set of slave rollers **104**. However, due to the rotation of the second independent

pulley **116B**, the package **130** can be shifted from the second set of slave rollers **106** towards the subsequent conveyor system **402**, while each slave roller from the first set of slave rollers **104** remains in steady state and does not rotate.

[0104] The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the order of steps in the foregoing embodiments can be performed in any order. Words such as “thereafter,” “then,” “next,” etc. are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Further, any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

[0105] It is to be appreciated that “one or more” includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

[0106] Moreover, it will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first frame could be termed a second frame, and, similarly, a second frame could be termed a first frame, without departing from the scope of the various described embodiments. The first frame and the second frame are both frames, but they are not the same contact.

[0107] The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or

“comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0108] As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

[0109] The systems, apparatuses, devices, and methods disclosed herein are described in detail by way of examples and with reference to the figures. The examples discussed herein are examples only and are provided to assist in the explanation of the apparatuses, devices, systems, and methods described herein. None of the features or components shown in the drawings or discussed below should be taken as mandatory for any specific implementation of any of these the apparatuses, devices, systems or methods unless specifically designated as mandatory. For ease of reading and clarity, certain components, modules, or methods may be described solely in connection with a specific figure. In this disclosure, any identification of specific techniques, arrangements, etc. are either related to a specific example presented or are merely a general description of such a technique, arrangement, etc. Identifications of specific details or examples are not intended to be, and should not be, construed as mandatory or limiting unless specifically designated as such. Any failure to specifically describe a combination or sub-combination of components should not be understood as an indication that any combination or sub-combination is not possible. It will be

appreciated that modifications to disclosed and described examples, arrangements, configurations, components, elements, apparatuses, devices, systems, methods, etc. can be made and may be desired for a specific application. Also, for any methods described, regardless of whether the method is described in conjunction with a flow diagram, it should be understood that unless otherwise specified or required by context, any explicit or implicit ordering of steps performed in the execution of a method does not imply that those steps must be performed in the order presented but instead may be performed in a different order or in parallel.

[0110] Throughout this disclosure, references to components or modules generally refer to items that logically can be grouped together to perform a function or group of related functions. Like reference numerals are generally intended to refer to the same or similar components. Components and modules can be implemented in software, hardware, or a combination of software and hardware. The term “software” is used expansively to include not only executable code, for example machine-executable or machine-interpretable instructions, but also data structures, data stores and computing instructions stored in any suitable electronic format, including firmware, and embedded software. The terms “information” and “data” are used expansively and includes a wide variety of electronic information, including executable code; content such as text, video data, and audio data, among others; and various codes or flags. The terms “information,” “data,” and “content” are sometimes used interchangeably when permitted by context.

[0111] The hardware used to implement the various illustrative logics, logical blocks, modules, and circuits described in connection with the aspects disclosed herein can include a general purpose processor, a digital signal processor (DSP), a special-purpose processor such as an application specific integrated circuit (ASIC) or a field programmable gate array (FPGA), a programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor can be a microprocessor, but, in the alternative, the processor can be any processor, controller, microcontroller, or state machine. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Alternatively, or in addition, some steps or methods can be performed by circuitry that is specific to a given function.

[0112] In one or more example embodiments, the functions described herein can be implemented by special-purpose hardware or a combination of hardware programmed by firmware or other software. In implementations relying on firmware or other software, the functions can be performed as a result of execution of one or more instructions stored on one or more non-transitory computer-readable media and/or one or more non-transitory processor-readable media. These instructions can be embodied by one or more processor-executable software modules that reside on the one or more non-transitory computer-readable or processor-readable storage media. Non-transitory computer-readable or processor-readable storage media can in this regard comprise any storage media that can be accessed by a computer or a processor. By way of example but not limitation, such non-transitory computer-readable or processor-readable media can include random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, disk storage, magnetic storage devices, or the like. Disk storage, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc™, or other storage devices that store data magnetically or optically with lasers. Combinations of the above types of media are also included within the scope of the terms non-transitory computer-readable and processor-readable media. Additionally, any combination of instructions stored on the one or more non-transitory processor-readable or computer-readable media can be referred to herein as a computer program package.

[0113] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of teachings

presented in the foregoing descriptions and the associated drawings. Although the figures only show certain components of the apparatus and systems described herein, it is understood that various other components can be used in conjunction with the supply management system. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, the steps in the method described above can not necessarily occur in the order depicted in the accompanying diagrams, and in some cases one or more of the steps depicted can occur substantially simultaneously, or additional steps can be involved. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

[0114] It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the disclosure being indicated by the following claims.

Claims

1. A conveyor system comprising: a first set of slave rollers defining a first conveying section; a second set of slave rollers defining a second conveying section separate from the first conveying section; and, a motor driven roller (MDR) comprising at least a fixed pulley, a first independent pulley, and a second independent pulley, wherein the first independent pulley is coupled with the first set of slave rollers and the second independent pulley is coupled with the second set of slave rollers, and wherein the motor driven roller is configured to drive at least one of the first set of slave rollers or the second set of slave rollers based on selective engagement of at least one of the first independent pulley or the second independent pulley.
2. The conveyor system of claim 1, wherein the first independent pulley is coupled with each slave roller from the first set of slave rollers through a first set of belts and the second independent pulley is coupled with each slave roller from the second set of slave rollers through a second set of belts.
3. The conveyor system of claim 1, wherein the fixed pulley comprises at least a first bearing hub, a first bearing, and a rotating shaft.
4. The conveyor system of claim 2, wherein the first independent pulley comprises at least a first electromagnet clutch, a first electromagnet, a second bearing, and a second bearing hub.
5. The conveyor system of claim 2, wherein the second independent pulley comprises a second electromagnet clutch, a second electromagnet, a third bearing, and a third bearing hub.
6. The conveyor system of claim 3, further connected to a controller, wherein, the controller is configured to receive an upstream photo eye signal from an upstream photo eye sensor based on presence of a package in proximity of the first set of slave rollers and in response to the upstream photo eye signal, the controller is further configured to drive the motor driven roller through the fixed pulley.
7. The conveyor system of claim 6, wherein, the controller is further configured to control the motor driven roller to activate a first electromagnet of the first independent pulley, wherein the first electromagnet is activated to selectively engage the first independent pulley with the rotating shaft of the motor driven roller through a first electromagnet clutch and to drive each slave roller from the first set of slave rollers through a second bearing, a second bearing hub, and the first set of belts.
8. The conveyor system of claim 6, wherein the motor driven roller comprises an inbuilt motor and the controller is configured to drive the motor driven roller using the inbuilt motor and through the fixed pulley.
9. The conveyor system of claim 6, wherein the motor driven roller is connected to an external motor and the controller is configured to drive the motor driven roller using the external motor and through the fixed pulley.
10. The conveyor system of claim 7, wherein, the controller is further configured to receive a first

photo eye signal from a first photo eye sensor based on presence of the package on the first set of slave rollers and in response to the first photo eye signal, control the motor driven roller to activate a second electromagnet of the second independent pulley, wherein the second electromagnet is activated to selectively engage the second independent pulley with the rotating shaft of the motor driven roller through a second electromagnet clutch and to drive each slave roller from the second set of slave rollers through a third bearing, a third bearing hub, and the second set of belts.

11. The conveyor system of claim 10, wherein, the controller is further configured to receive a second photo eye signal from a second photo eye sensor based on the presence of the package on the second set of slave rollers.

12. The conveyor system of claim 11, wherein, the controller is further configured to determine, through the first photo eye sensor, if the package is present on the first set of slave rollers and in response to presence of the package, control the motor driven roller to continue engagement of the first independent pulley with the rotating shaft of the motor driven roller to drive each slave roller from the first set of slave rollers.

13. A method of controlling a conveyor system, comprising: defining a first conveying section by a first set of slave rollers; defining a second conveying section by a second set of slave rollers, wherein the second conveying section is defined separately from the first conveying section; arranging a motor driven roller between the first set of slave rollers and the second set of slave rollers, wherein the motor driven roller comprising at least a fixed pulley, a first independent pulley, and a second independent pulley; coupling the first independent pulley of the motor driven roller with the first set of slave rollers; coupling the second independent pulley of the motor driven roller with the second set of slave rollers; driving, through the motor driven roller, the first set of slave rollers based on selective engagement of the first independent pulley; and, driving, through the motor driven roller, the second set of slave rollers based on selective engagement of the second independent pulley.

14. The method of claim 13, further comprising, coupling the first independent pulley with each slave roller from the first set of slave rollers through a first set of belts and coupling the second independent pulley with each slave roller from the second set of slave rollers through a second set of belts.

15. The method of claim 13, further comprising, receiving an upstream photo eye signal from an upstream photo eye sensor based on presence of a package in proximity of the first set of slave rollers and in response to the upstream photo eye signal, driving the motor driven roller through the fixed pulley.

16. The method of claim 15, further comprising, controlling the motor driven roller for activating a first electromagnet of the first independent pulley, wherein the first electromagnet is activated for selectively engaging the first independent pulley with a rotating shaft of the motor driven roller through a first electromagnet clutch and for driving each slave roller from the first set of slave rollers through a second bearing, a second bearing hub, and the first set of belts.

17. The method of claim 15, further comprising, receiving a first photo eye signal from a first photo eye sensor based on the presence of the package on the first set of slave rollers and in response to the first photo eye signal, controlling the motor driven roller for activating a second electromagnet of the second independent pulley, wherein the second electromagnet is activated for selectively engaging the second independent pulley with a rotating shaft of the motor driven roller through a second electromagnet clutch for driving each slave roller from the second set of slave rollers through a third bearing, a third bearing hub, and the second set of belts.

18. The method of claim 17, further comprising, determining, through the first photo eye sensor, if the package is present on the first set of slave rollers and in response to presence of the package, controlling the motor driven roller to continue engagement of the first independent pulley with the rotating shaft of the motor driven roller for driving each slave roller from the first set of slave rollers.

19. The method of claim 17, further comprising, determining, through the first photo eye sensor, if the package is not present on the first set of slave rollers and in response to absence of the package, controlling the motor driven roller for deactivating a first electromagnet of the first independent pulley, wherein the first electromagnet is deactivated for disengaging the first independent pulley from the rotating shaft of the motor driven roller to stop driving each slave roller from the first set of slave rollers.

20. A pulley for a roller of a conveyor system, wherein the pulley comprising: a fixed pulley comprising a first surface, a second surface, a first bearing hub, a first bearing, and a rotating shaft, wherein the first surface of the fixed pulley is configured to function as a first edge of the pulley; a first independent pulley comprising a third surface, a fourth surface, a first electromagnet clutch, a first electromagnet, a second bearing, and a second bearing hub, wherein the third surface of the first independent pulley is coupled with the second surface of the fixed pulley; and, a second independent pulley comprising a fifth surface, a sixth surface, a second electromagnet clutch, a second electromagnet, a third bearing, and a third bearing hub, wherein the fifth surface of the second independent pulley is coupled with the fourth surface of the first independent pulley and the sixth surface of the second independent pulley is configured to function as a second edge of the pulley.
